

CoRTeC: Cohort-Conditioned Differentially Private Synthetic Tabular Data from a Frozen Language Model

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Abstract. Differentially private (DP) synthetic tabular data lets an institution train on, and share, records it cannot release. Its value depends on how much accuracy a model loses when it is trained on the synthetic data instead of the real data. The deployed mechanisms, MST and AIM, are selected on marginal fidelity and do not report it. On UCI Adult at $\epsilon = 2$ their output trains models 0.14 to 0.17 AUC below a same-size real sample, and real data with its target permuted passes a 90% marginal-similarity bar. CoRTeC spends the privacy budget once, on a release built for a downstream model, and trains nothing. Within public cohorts it publishes one DP histogram per attribute and outcome class, and a DP table of target rates over disjoint cells. Disjoint cells cost one query per level, so the release carries $6.7\times$ more budget per statistic. A frozen language model then decodes the release: each batch is told the exact counts it must produce, a threefold pool is filtered to the rows matching the release, and sub-bin values are redrawn. The generator never sees a private record, so generation is post-processing: unlimited datasets at no further privacy cost. A third stage releases a DP bound on the gap between private and synthetic conditional rates. At $\epsilon = 2$ and $n = 300$ on Adult, credit and NHANES, CoRTeC's 1-way error is within 0.005 of MST's or below it, and below a real sample's. Tree models trained on its output match those trained on a same-size real sample on all three datasets; the linear model matches on Adult and is within 0.02 elsewhere; AIM's and MST's output trains models 0.05 to 0.18 AUC lower. Whether a forced private relationship survives synthesis depends on the mechanism family, not the guarantee. Enabling reasoning in the generator changes conditional error $3.8\times$ while AUC cannot detect it. Four membership-inference attacks reach an advantage of 0.048 against a permitted 0.762.

1. Introduction

An institution that holds sensitive tabular data usually wants three things its obligations forbid. It wants to train a model outside its own perimeter; to give a realistic sample to a partner or a vendor; and to let researchers work on the problem without working on the people. Synthetic data with a differential privacy guarantee [13, 14] is the accepted way to do all three. Whether it is worth doing depends on one question: if I train on the synthetic data instead of the real data, how much accuracy do I lose?

The mechanisms an institution can deploy today do not answer that question in their own papers. The marginal-based synthesisers MST [28] and AIM [29] are evaluated on marginal workload error, the error in the distributions of single columns and of small groups of columns. They attain low error on that measure, and they were built for published tables and contingency reports rather than for training models.

The generative models trained under differential privacy [24, 38, 49] are evaluated on downstream utility, the accuracy of a model trained on the synthetic data and tested on real data. At the budgets and schema sizes we test, they do not preserve the target's base rate.

The methods that do reach real-data utility fine-tune a language model on the records [5, 45]. They either carry no privacy guarantee or train on the private data under DP-SGD [1], gradient descent with clipped and noised updates. DP-SGD is expensive, and it places the private data on the training hardware. PATE [35, 36], an ensemble of teachers trained on disjoint shards of the private data whose noised votes label a public dataset, would avoid that. But it needs a public transfer set drawn from roughly the same distribution as the private data, and where privacy matters most, in a hospital's encounter records or a bank's default history, no such set exists.

A practitioner therefore chooses between two options. One is a mechanism that optimises low-order marginals rather than the structure a downstream model needs. The other is a large language model asked to invent plausible records. That is fast and needs no privacy accounting, but the model knows nothing about the institution's data.

CoRTeC. CoRTeC separates two things every other method does together: spending the privacy budget and producing records. First it spends the budget, once, on a statistics release. The release contains:

- cohorts (groups of records) formed by a public stratification rule, meaning bands of one or two attributes fixed before the data is seen;
- one differentially private histogram per attribute and per outcome class inside each cohort;

- a private class balance (the share of each outcome) and a private cohort size; and
- a private table of target rates over the cells of a conditional hierarchy, meaning nested subgroups defined by one, two or three attributes, each subgroup being a cell.

Then a frozen, un-finetuned language model reads that release, and nothing else, and writes records that match it. Each batch is told the exact number of rows it must produce per bin, per category and per outcome. From a pool of generated rows, the pipeline keeps the rows whose cell counts match the release. Wherever the release describes the class shape at bin resolution, the kept values are redrawn inside their released bins.

This design has four consequences, and they are the reasons to use it.

The privacy argument is small. The generator never sees a private record, so its work is post-processing of a differentially private release: a computation on the release alone, which adds no privacy cost. The privacy analysis therefore reduces to the release, which is a composition of counting queries.

Generation costs no privacy and can run anywhere. Any number of datasets can be drawn from one release at no further privacy cost, and the model can be a tenant-isolated enterprise endpoint or an air-gapped local model with the same guarantee.

The budget goes to the structure a classifier needs, and it goes there cheaply. A downstream model needs the conditional structure and the per-class shape of each attribute. Histograms have sensitivity 1 regardless of resolution, and disjoint cohorts and cells compose in parallel, so the release carries that structure at 6.7× more budget per statistic than a natural implementation.

No statistician is needed. An auto-configurator chooses the cohorts and the hierarchy from the schema and the budget.

What we find. We test at $\epsilon = 2$, where ϵ is the privacy budget and smaller is stronger, and at $n = 300$, on three datasets: census, finance, and a national health examination survey.

On fidelity, CoRTeC’s 1-way marginal error, the error in each single column’s distribution, is within 0.005 of MST’s on Adult and finance, and below it on NHANES. It is below the error of a real sample of the same size, and below AIM’s, on every dataset. Its 2-way error, over pairs of columns, is below both marginal methods’ on every dataset.

On utility, tree models trained on CoRTeC’s output match the real-sample floor on every dataset. The floor is the accuracy of a model trained on a real sample of the same size. The linear model matches it on Adult and is within 0.02 of it on finance and NHANES. Models trained on AIM’s and MST’s output score 0.05 to 0.18 lower in AUC, the area under the ROC curve, which is the utility measure used throughout. Of 35 comparisons against AIM and MST on Adult and finance, corrected for multiple testing with Holm’s procedure, 28 favour CoRTeC, one favours a baseline, MST’s 0.005 advantage on 1-way error on Adult, and six do not separate. AIM keeps the lowest error on its own 3-way workloads, sets of three-column marginals, at adequate sample size, which its design predicts.

Two results corrected our own framing. A private relationship forced into the data survives MST essentially perfectly (slope 0.998), so the marginal methods’ utility gap is not an inability to carry conditional structure. Their loss is measurable instead as a 2.2 to 2.8× inflation of pairwise feature dependence relative to real data of the same size; CoRTeC inflates it by 1.24×.

The second result is that CoRTeC’s own conditional fidelity, how closely the outcome rate within each subgroup matches the private data, is decided by whether reasoning is enabled in the generator, the model’s option to think through a request before answering. That single flag changes conditional error by a factor of 3.8, and downstream AUC does not detect the change. This finding has a methodological consequence, developed in Section 6.3: both criteria the field selects synthesisers on, marginal similarity and aggregate downstream utility, are saturated. A known-useless dataset passes the first, and on a public benchmark an ungrounded language model passes the second.

Contributions. In summary, this paper makes the following contributions.

- A privacy-budget allocation that makes conditional structure cheap. Public stratification costs nothing; histograms replace moment queries; one histogram block per outcome class costs what one pooled block does; and parallel composition, under which statistics on non-overlapping groups each spend the full budget rather than sharing it, applies across disjoint cohorts and cells. The result is 6.7× more budget per released statistic at unchanged ϵ (Section 4).

- A decoding stage that reaches the release’s own fidelity without spending budget: exact-count batches, selection from a threefold pool by raking and refinement over every released cell, and a sub-bin rule that makes values inside released bins the release’s own (Sections 4.4 and 4.5).
- A head-to-head against the deployed mechanisms that reports fidelity and utility in the same table, against a real-sample floor and a no-information floor (a real sample of the same size, and the same data with its target permuted), with Holm-corrected families of tests (Section 6.1).
- Evidence that the two standard acceptance criteria are saturated, a concrete replacement criterion, and a matched control that separates a mechanism reading a release from one reciting its prior (Section 6.3).
- A transmission test, which asks whether a relationship planted in the private data reappears in the synthetic output, showing that whether a private conditional relationship survives synthesis is a property of the mechanism family and not of the guarantee (Section 6.4).
- A utility transmission bound (Stage C): a budgeted, auditable bound on the gap between private and synthetic conditional rates, with floors and a ceiling built in and a refusal to report when it cannot discriminate (Sections 4.6 and 6.9).
- Auto-configuration; a membership-inference evaluation, in which attacks that try to tell whether a record was in the private data are validated on a positive control of real records passed off as synthetic; a reference deployment architecture with the trust boundary located precisely; a catalogue of the measurement artefacts that produced wrong conclusions along the way; and two Apache-2.0 implementations that enforce every guarantee in code (Sections 6.7, 6.9, 6.11, 7 and 9).

A companion [technical report](#) contains every intermediate experiment, the per-dataset walkthroughs and the full defect catalogue. This paper reports the final mechanism and the evidence for it.

2. Background

2.1 Data and histograms

Definition 1 (Dataset). A dataset D is a multiset of n records over a schema with attributes $A = A_{num} \cup A_{cat}$ and a binary target y . Each attribute a has a public domain, an interval $[lo_a, hi_a]$ if numerical and a finite value set if categorical, and public bin edges E_a partition each numerical domain into intervals.

Definition 2 (Histogram and rate). $M_a(D)$ is the histogram of attribute a over E_a , or over its value set if a is categorical: a vector of counts, one per bin. For a cell c of the record space, $k_c(D)$ is the number of records in c with $y = 1$, and $\rho_c(D) = k_c(D) / |c|$ is the target rate in c .

A partition Π of the record space induced by a rule over public domains assigns every record to one part, $\Pi(x)$, as a function of x alone. Two properties of the schema are assumed public throughout, as in every marginal-based synthesiser: the attribute list and each attribute’s domain bounds. No quantity derived from private values is used to choose bins, bands or cells. The one exception is the auto-configurator’s column ranking, which is charged to the budget (Section 4.3).

2.2 Differential privacy

Two datasets D, D' are neighbours if one is obtained from the other by adding or removing one record.

Definition 3 (Differential privacy [13]). A randomised mechanism M satisfies ϵ -differential privacy if for all neighbouring D, D' and all sets S of outputs,

$$\Pr[M(D) \in S] \leq e^\epsilon \cdot \Pr[M(D') \in S]. \quad (1)$$

Proposition 1 (Laplace mechanism [13]). Let f be a function of D with L1 sensitivity $\Delta = \max_{D, D'} \|f(D) - f(D')\|_1$. Then $f(D) + \text{Lap}(\Delta/\epsilon)$, with the noise drawn independently per coordinate, satisfies ϵ -differential privacy.

A histogram over any number of bins has $\Delta = 1$, because one record moves exactly one count by one. A counting query has $\Delta = 1$ for the same reason. The mechanism is built on this fact: an entire distributional shape costs the same budget as a single count.

Proposition 2 (Sequential composition [14]). If $M_1, \dots, M_k$ satisfy $\epsilon_1, \dots, \epsilon_k$ -differential privacy and are run on the same data, their joint output satisfies $(\sum_i \epsilon_i)$ -differential privacy.

Proposition 3 (Parallel composition [30]). *If $D_1, \dots, D_k$ are disjoint subsets of D defined by a data-independent rule and each M_i satisfies ϵ -differential privacy on D_i , the joint output satisfies ϵ -differential privacy.*

Proposition 4 (Post-processing [14]). *If M satisfies ϵ -differential privacy and g is any function, possibly randomised, that does not read D , then $g \circ M$ satisfies ϵ -differential privacy.*

Definition 3 also bounds any membership adversary. No test that decides whether a record is in D can have a true-positive rate exceeding its false-positive rate, which is its advantage, by more than

$$\frac{e^\epsilon - 1}{e^\epsilon + 1}, \quad (2)$$

which is 0.762 at $\epsilon = 2$. We use this bound as the reference for the measured advantages of Section 6.11.

2.3 Related work

Marginal and graphical-model methods. PrivBayes [52], MST [28], PrivMRF [7] and AIM [29] select low-order marginals under differential privacy and fit a graphical model with Private-PGM [27]. MST won the 2018 NIST differential privacy synthetic data challenge [37]. AIM is workload-adaptive and is the strongest of these.

Both MST and AIM are evaluated on marginal workload error. MST uses the NIST score at $\epsilon \in \{0.3, 1, 8\}$; AIM uses normalised L1 workload error over all-3-way, target and skewed workloads. Neither paper reports a downstream predictive-utility experiment. We state this as a scope observation rather than a criticism: these mechanisms target a different objective and attain it, and Section 6.1 shows them doing so.

The benchmark of Tao et al. [44], which does include a classification task, found the marginal methods ahead of the generative ones, and our Tables 3 to 5 reproduce that result for the baselines they share. A later benchmark by Chen et al. [10] reaches the same conclusion with a machine-learning utility score. It excludes language-model-based methods on the ground that their pretraining corpora would bias the evaluation, an objection Section 6.3 is designed to answer.

Generative models trained under differential privacy. DP-GAN [49], PATE-GAN [24], and DP-CTGAN and PATE-CTGAN [38] apply DP-SGD or PATE [35, 36] to a generative model, the last two over the CTGAN architecture [50]. Rosenblatt et al. [38] evaluate downstream utility with train-on-synthetic, test-on-real (TSTR) against train-on-real. We adopt their protocol.

Language models for tabular data. GReaT [5] fine-tunes a language model to emit rows and achieves high fidelity with no privacy guarantee, since the model memorises records. DP-LLMTGen [45] adds DP-SGD fine-tuning, which is private but expensive and still trains on the private data. DP-2Stage [3] reduces the cost of that fine-tuning by first training on public pseudo-data, but still trains on the private records. Curated LLM [39] uses a language model to augment small tabular datasets without a privacy guarantee, and conditions on the private data directly rather than on a release.

Private synthesis through model APIs. The closest line of work to ours also keeps the foundation model frozen and reaches it only through an inference API. Private Evolution [26, 48] generates candidate records from the model, privately votes on which candidates lie nearest to the private records, and iterates; it spends privacy budget on every round of votes. Swanberg et al. [43] adapt it to tabular data with a workload-based distance and find that API access to a strong model does not by itself improve on the marginal baselines. Tran et al. [46] extend it with evolutionary operators and private scoring. Their one-shot variants treat the model’s schema-only output as public data and spend the budget reweighting it to privately measured marginals, which is the nearest analogue to our selection step. The difference is that our release, not a later measurement, is what the model decodes, and the selection reads only that release.

CoRTeC differs in where the budget goes and how often it is spent. It spends the budget once, on a statistics release that names the conditional structure a downstream model needs, and then treats the model as a decoder of that release under exact counts and selection. The model is therefore never in the privacy loop; any number of datasets can be drawn from one release; and the privacy analysis is a composition argument over counting queries rather than over rounds of private selection. Table 1 summarises the positioning.

Table 1: Positioning. Every prior method lets the generator see private data, and privacy depends on the training mechanism. CoRTeC trains nothing: it releases private statistics once and lets a frozen model decode them, so the privacy argument is a post-processing argument over the release. The trade-off is explicit. Any structure the released statistics do not imply comes from the model’s prior, which is the failure mode Section 6.4 is designed to detect.

method family	generator sees raw data?	where noise enters	utility in its own paper?	conditional structure
PrivBayes, MST, AIM, PrivMRF	yes (to fit)	selected marginals	no	only what the chosen marginals imply
DP-GAN, DP-CTGAN	yes (DP-SGD)	gradients	partially	learned, degraded by clipping and noise
PATE-GAN, PATE-CTGAN	yes (PATE on the discriminator)	teacher votes	yes	learned
GReaT	yes (fine-tuning)	none; not private	yes	learned, memorised
DP-LLMTGen, DP-2Stage	yes (DP-SGD fine-tuning)	gradients	yes	learned
Private Evolution (API-only)	no (private votes over candidates)	each round's nearest-neighbour votes	yes	whatever the votes select; budget spent per round
CoRTeC (this work)	no	released statistics only	yes	released explicitly: a private conditional table and one histogram block per outcome class

3. Problem Setting and Threat Model

The task. Given a private dataset D with a binary target, a public schema and a budget ϵ_{total} , produce a synthetic dataset $\hat{D}$ of any requested size such that:

- (i) the whole procedure is ϵ_{total} -differentially private with respect to D ; and
- (ii) a model trained on $\hat{D}$ performs, on real held-out data (records no mechanism saw), as a model trained on a real sample of the same size would.

We measure (ii) directly, as TSTR under three student models, the classifiers trained on the synthetic data, against a matched real sample. We always measure fidelity beside it, because a method can score well on one and poorly on the other.

The trust boundary. Figure 1 shows where it lies. Everything to the left of the boundary runs inside the institution's regulated zone and touches private records: the extract, the schema declaration and the release engine. The only object that crosses the boundary is the release R , a set of noisy counts. The generator, the selection step, the evaluation and every downstream use run to the right of the boundary, and read R and their own output only.

Whoever operates the generator, including a hostile endpoint, learns nothing that R does not already reveal, and R is ϵ_{total} -differentially private. This is what lets the model be a vendor-hosted endpoint: the privacy argument holds against a hostile endpoint by construction, because the request carries only R . What does not hold against a hostile endpoint is the compliance argument, which rests on tenant isolation, private networking, data residency and a business associate agreement. Section 7 states those requirements.

The adversary. For the guarantee, the adversary is unrestricted: Definition 3 bounds any function of the release and of everything derived from it. For the empirical evaluation of Section 6.11, the adversary sees $\hat{D}$ and a candidate record and must decide whether the record was in D . That measurement tests the implementation, not the mechanism, and cannot improve on the bound (2).

The privacy unit. ϵ protects a row. On data with one row per person, $\epsilon_{\text{person}} = \epsilon_{\text{row}}$, which is the case for Adult, the credit dataset, NHANES and every other dataset we use except one. On encounter-level data with repeated individuals, group privacy weakens the per-person guarantee by the maximum number of rows one person contributes. Diabetes 130 has a mean of 1.42 rows per patient and a maximum of 40, so at $\epsilon_{\text{row}} = 2$ its worst-case per-person guarantee is 80, outside any range normally considered meaningful. This is equally true of MST or AIM run on the same data at the same row-level ϵ . Every figure we report from that dataset is a row-level figure.

The mitigation is a pre-processing step taken before the release, and it is exact. Capping each person's contribution to their first C rows discards data and does not change the quantity being estimated. On Diabetes 130 at a matched $\epsilon_{\text{person}} = 2$ it is 6.3 $\times$ more accurate than aggregating to one row per person (conditional error 0.0123 against 0.0780). The reference implementations require the schema to declare the maximum rows per person, and refuse a declared unit whose ϵ_{person} is vacuous unless the caller acknowledges it in the audit trail.

What the guarantee does not cover. The frozen model’s pretraining corpus is not part of it. If a private record was in that corpus, it was compromised before CoRTeC ran. Language models are known to reproduce training data even under benign prompting [2]. What CoRTeC guarantees is that the generator learns nothing about D from CoRTeC; it cannot guarantee what the generator already knew. Section 6.3 measures the one consequence that would inflate our results, a prior that happens to be right about a public benchmark. Section 6.11 attacks the artefact the method publishes, which is where a memorised record would have to surface to cause harm.

4. The CoRTeC Mechanism

4.1 Overview

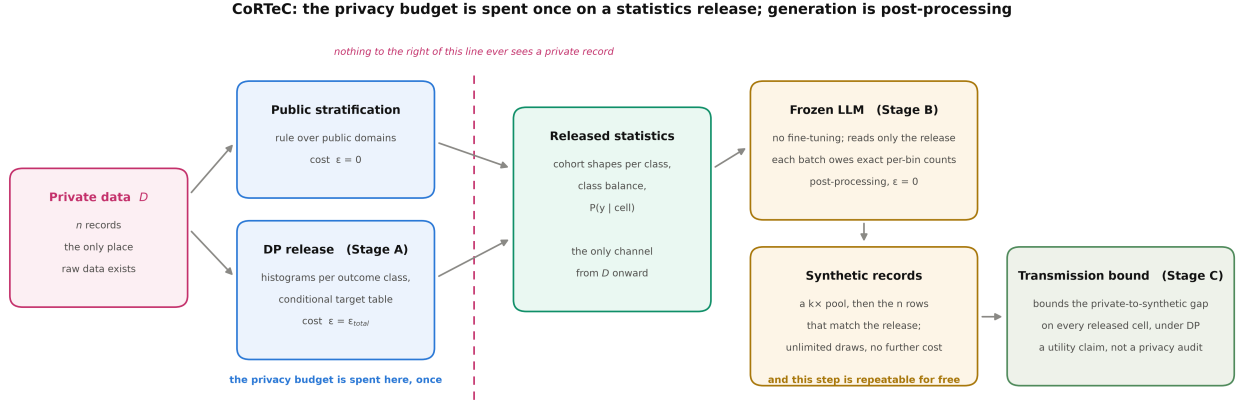


Figure 1: CoRTeC. The privacy budget is spent once, in Stage A. Everything to the right of the dashed line, the generator and every post-processing step, sees only released statistics, so it contributes nothing to the privacy cost and may be repeated without limit.

The mechanism has three stages.

Stage A (Algorithm 1) is the only code that reads D . It partitions the record space into cohorts by a public rule. Inside each cohort of at least $n_{\min}$ records it releases a noisy cohort size, a noisy class balance, and one noisy histogram per attribute and per outcome class. It then releases, for each level of a nested conditional hierarchy, a noisy positive count and a noisy size for every cell of at least $n_{\min}$ records. The output is the release R .

Stage B reads R and nothing else. It:

- allocates rows to cohorts by released size;
- apportions exact per-bin counts from the released histograms;
- asks a frozen language model for batches of rows that carry exactly the counts still owed (Algorithm 3);
- generates a pool k times larger than the requested size and keeps the m rows whose cell counts best match R (Algorithm 4); and
- redraws each kept value uniformly inside its released bin wherever the release describes the class shape at that resolution.

Stage C (Algorithm 5) is optional and spends a separately declared budget. It releases a Laplace estimate of each cell’s private target rate and turns it into a simultaneous bound on how far the synthetic data’s rates lie from the private ones. A deployment can then attach a quantified utility claim to a release package without exposing the private data.

An auto-configurator (Algorithm 2) derives the cohort rule and the conditional hierarchy from the schema, the budget and the requested size, and charges the one step that reads private data.

4.2 Stage A: the release

Algorithm 1: CoRTeC-Release

Input: private D ; public schema (domains, bin edges E , stratification rule Π , conditional levels $L_0 \subset L_1 \subset \dots \subset L_k$); budget ϵ_{total} ; conditional share α ; count share γ ; suppression floor $n_{\min}$

Output: release R

- (1) $\epsilon_{\text{count}} \leftarrow \gamma \cdot \epsilon_{\text{total}}$; $\epsilon_{\text{rest}} \leftarrow \epsilon_{\text{total}} - \epsilon_{\text{count}}$; $\epsilon_{\text{marg}} \leftarrow (1 - \alpha) \cdot \epsilon_{\text{rest}}$; $\epsilon_{\text{cond}} \leftarrow \alpha \cdot \epsilon_{\text{rest}}$
 - (2) $q \leftarrow |A_{\text{num}}| + |A_{\text{cat}}| + 1$ (queries per cohort; sequential within a cohort)
 - (3) $\epsilon_q \leftarrow \epsilon_{\text{marg}} / q$ (cohorts are disjoint: parallel across Π)
 - (4) $\epsilon_L \leftarrow \epsilon_{\text{cond}} / |L_{\text{viable}}|$ (levels overlap: sequential across levels)
 - (5) For each cohort $P \in \Pi$ with $|P| \geq n_{\min}$:
 - (6) Release $n_p^* \leftarrow \max(n_{\min}, |P| + \text{Lap}(1/\epsilon_{\text{count}}))$
 - (7) Release the two-bin count histogram of y on P , each count $+ \text{Lap}(1/\epsilon_q)$
 - (8) If $|P \cap \{y = 1\}| \geq n_{\min}$ and $|P \cap \{y = 0\}| \geq n_{\min}$: (class-conditional blocks)
 - (9) For $c \in \{0, 1\}$ and $a \in A$: release $M_a^*(P, c) \leftarrow M_a(P \cap \{y = c\}) + \text{Lap}(1/\epsilon_q)$
 - (10) $M_a^*(P) \leftarrow$ mixture of $M_a^*(P, c)$ under the released class balance
 - (11) Else: for $a \in A$, release $M_a^*(P) \leftarrow M_a(P) + \text{Lap}(1/\epsilon_q)$ (pooled block)
 - (12) For each level $\ell \in L_{\text{viable}}$ and each cell $c \in \ell$ with $|c| \geq n_{\min}$:
 - (13) Release $k_c^* \leftarrow k_c(D) + \text{Lap}(2/\epsilon_L)$ and $n_c^* \leftarrow |c| + \text{Lap}(2/\epsilon_L)$
 - (14) $\rho_c^* \leftarrow \text{clip}(k_c^* / \max(n_c^*, n_{\min}), 0, 1)$ (post-processing of two counts)
 - (15) Return R : every released quantity, with its ϵ , sensitivity and composition rule
-

Five decisions in Algorithm 1 make the method work, and each rests on a sensitivity argument.

Public stratification (line 5). Cohorts come from a rule fixed over public domains, so $\Pi(x)$ is a deterministic function of x 's own record, and forming cohorts consumes no budget. This replaces private clustering, which spends budget and, in our measurements, produced degenerate cohorts of 3 and 8 records.

Histograms, not moments (lines 9 and 11). A histogram has L1 sensitivity 1 regardless of its bin count, since one individual moves exactly one bin. An entire distributional shape therefore costs the same budget as a single count, and moments are recovered from the released histogram for free. Under the moment formulation, by contrast, the standard-deviation query has sensitivity $(hi - lo)/(2\sqrt{n})$, and at realistic budgets that produced released values larger than the attribute's entire range: at a cohort size of 20, a released standard deviation of 693.80 against a true value near 13.

Class-conditional histograms (lines 8 to 10). A pooled cohort histogram says what the cohort looks like, and nothing about how any feature relates to the target outside the conditional hierarchy. The generator fills that in from its prior, and on the finance dataset the filled-in columns measurably degraded the downstream model (Section 6.2). Partitioning each cohort by the target, and releasing one histogram block per (cohort, class), gives the generator the class-conditional marginals a classifier needs, for every column. The two class blocks are disjoint, so they compose in parallel, and each spends the same ϵ_q per column that the pooled block did. The pooled histogram is their mixture under the released class balance, which is post-processing. Where a class falls below $n_{\min}$, the cohort keeps the pooled block.

A conditional table over a disjoint partition (lines 12 to 14). Because the cells of one level partition D , they satisfy parallel composition: each receives the full ϵ_L , however many cells the level contains. A cell's rate is the ratio of two counts, each carrying $\text{Lap}(2/\epsilon_L)$. Its noise is therefore of order $2/(|c| \cdot \epsilon_L)$, and its accuracy is governed by cell support, not by the number of cells. Conditional structure is therefore cheap to release, and that observation is the basis of the method.

We do not noise the rate directly as a bounded mean. That mechanism's scale, $1/(|c| \cdot \epsilon_L)$, depends on $|c|$, which is itself private under add/remove adjacency. Two Laplace densities with different scales have an unbounded likelihood ratio in one tail, so the mechanism is not pure ϵ -differentially private. Two counts of sensitivity 1 at a public scale are.

The published cohort size (line 6). The generator allocates rows to cohorts in proportion to size, so the size must be released. It is a private count, noised at $\text{Lap}(1/\epsilon_{\text{count}})$, charged out of the marginal share and clamped at $n_{\min}$. A cohort is released only when it holds at least $n_{\min}$ records, so any lower published value is impossible under the release’s own rule. Without the clamp, at $\epsilon = 0.3$ on NHANES the youngest age band’s noised size clipped to zero, and the band vanished from the output while every per-row check passed.

The conditional share. With one histogram block per outcome class, the release already carries most of what a classifier needs. The conditional table’s share α can then fall from a half to a fifth without measurable loss. The histograms receive that budget, and the release’s own 1-way error against the training data falls from 0.029 to 0.020 on Adult and from 0.017 to 0.013 on finance. $\alpha = 0.2$ is the shipped default, and $\gamma = 0.02$.

4.3 Auto-configuration

Every published result in differentially private synthesis, and every early result in this project, depended on a hand-chosen configuration: which columns the conditional table conditions on, how many levels it carries, and how cohorts are formed. Choosing those columns after inspecting the private data is an unchanged, data-dependent decision, and it means the method cannot be deployed without a statistician. Algorithm 2 derives the whole configuration from the declared schema, the target, the budget and the number of records requested.

Algorithm 2: CoRTeC-Autoconfigure

Input: private D of size n ; public schema; target y ; ϵ_{total} ; requested size m ; noise-floor multiple M

Output: stratification rule Π , conditional levels L , budgets ϵ_{sel} and $\epsilon_{\text{release}}$

- (1) $m_{\text{cand}} \leftarrow$ number of candidate columns; $L \leftarrow$ median level count of the candidates; $|Y| \leftarrow$ number of target classes
 - (2) $f \leftarrow \text{clip}(M \cdot m_{\text{cand}} \cdot L \cdot |Y| / n, 0.05, 0.30)$; $\epsilon_{\text{sel}} \leftarrow f \cdot \epsilon_{\text{total}}$; $\epsilon_{\text{release}} \leftarrow \epsilon_{\text{total}} - \epsilon_{\text{sel}}$
 - (3) For each candidate column a : (the only step that reads D)
 - (4) $T_a \leftarrow$ contingency table of bands(a) $\times y$ on D , each cell $+ \text{Lap}(m_{\text{cand}}/\epsilon_{\text{sel}})$
 - (5) $\text{score}_a \leftarrow$ mutual information of T_a (post-processing)
 - (6) Coarsen a into at most four bands by its noisy positive rate, read from T_a
 - (7) Rank the candidates by score; choose the stratification column and the conditioning columns
 - (8) Add conditioning columns while every cell still expects at least 8 records under m , preferring the depth closest to 20 records per cell, divided by the cohort count
 - (9) Return $\Pi, L, \epsilon_{\text{sel}}, \epsilon_{\text{release}}$
-

Column ranking is the only private step, and it is charged. Each candidate is scored from one noised contingency table of its bands against the target. A record occupies exactly one cell, so the table’s L1 sensitivity is 1 however many cells it has. The tables are not disjoint across candidates, since every record appears in all of them, so they compose sequentially at $\epsilon_{\text{sel}}/m_{\text{cand}}$ each. Everything after the noised counts, namely the mutual-information computation, the coarsening and the ranking, is post-processing.

Greedy forward selection by conditional mutual information was implemented and reverted. A cardinality bias in that estimator rewards a wide candidate for having more cells, and on Diabetes 130 it lowered the release ceiling from 0.613 to 0.535 at every selection budget. Coarsening wide columns into at most four bands by their noisy positive rate reproduces what hand-tuned band objects did (education, 16 levels to 4) at no further cost.

Richness needs no private input at all. A conditional table’s usable resolution is set by how many records land in each cell; cell counts are a function of public cardinalities, and the number of records requested is a public choice.

The selection budget is adaptive because selection is meaningful only while a column’s expected cell count exceeds the Laplace noise on it. Writing f for the share of ϵ_{total} spent on selection, and requiring the expected count per cell to exceed the noise scale by a factor M ,

$$\frac{n}{L \cdot |Y|} \geq M \cdot \frac{m_{\text{cand}}}{f} \Leftrightarrow f \geq \frac{M \cdot m_{\text{cand}} \cdot L \cdot |Y|}{n}, \quad (3)$$

which is line 2 after clipping the share to $[0.05, 0.30]$. $M = 1$ is the shipped value, and every quantity in (3) except n comes from the declared schema. Small datasets therefore spend more on selection, which is correct: a smaller release needs less budget to describe. Under a fixed 5% share, three of six datasets it had never seen declined to configure at all (Section 6.7).

4.4 Stage B: generation with exact counts

Algorithm 3: CoRTeC-Generate

Input: release R ; requested size m ; pool factor k ; batch size B ; frozen model G

Output: pool $\hat{D}$ of about $k \cdot m$ rows

- (1) For each cohort P in R : $m_p \leftarrow k \cdot m \cdot n_p / \sum_p n_p$ (rows by released size)
 - (2) $T_p \leftarrow \text{Apportion}(m_p; \text{released class balance}, M_a(P, c) \text{ for all } a, c)$ (exact per-bin, per-category and per-class counts)
 - (3) While rows are still owed to P :
 - (4) $b \leftarrow \min(B, \text{rows owed})$
 - (5) $\text{owed} \leftarrow \text{Apportion}(b; T_p - \text{counts of the rows already accepted for } P)$
 - (6) $\text{rows} \leftarrow G(\text{prompt}(R, P, b, \text{owed}))$ (the prompt carries R and the counts, never D)
 - (7) Accept the rows that parse, lie inside their domains and carry declared categories; append them to $\hat{D}$
 - (8) Return $\hat{D}$
-

Asked to reproduce a histogram stated as shares, a frozen model reproduces it approximately, and the rows it emits carry that approximation plus the sampling noise of any batch. Asked instead for exact counts, it can satisfy them. A 25-row batch told the number of rows it must produce per bin and per category reproduces every count line exactly on the frontier models we measured; the same batch given shares matches about two thirds of them.

The counts are apportioned from the released histograms for the cohort as a whole (line 2), so they carry no more information than the release does. Each batch asks for the counts the cohort still owes after the rows already accepted (line 5), so a batch that returns more or fewer valid rows than asked cannot leave the cohort's totals short. The batch size is 25 and the pool factor is 3 by default.

The prompt (Appendix A) is versioned and hash-locked in the reference implementation, because two of its elements are essential and their removal is not visible in the output. The first is the conditional target table, which is the only channel through which the conditional structure reaches the generator. The second is the instruction to follow the released statistics even where they contradict the model's expectations; without it, a model reconciles the release against its prior and emits the prior. Section 6.3 measures both.

4.5 Selection, and values below bin resolution

Algorithm 4: CoRTeC-Select

Input: release R ; pool $\hat{D}$; requested size m

Output: synthetic dataset D^* of exactly m rows

- (1) Require that every cohort P holds at least $1.1 \cdot m \cdot n_p / \sum n$ rows of $\hat{D}$ (coverage guard)
 - (2) $w \leftarrow$ inclusion weights from iterative proportional fitting of $\hat{D}$ to every released cell: cohort sizes, class balances, one histogram per cohort, class and attribute; a category the release suppressed receives weight 0
 - (3) $D^* \leftarrow$ a systematic sample of m rows of $\hat{D}$ under w (rake)
 - (4) Repeat: swap a row of D^* for a row of $\hat{D} \setminus D^*$ whenever the swap lowers the weighted L1 distance between the cell counts of D^* and those of R , until no swap improves it (refine)
 - (5) For each cohort P with class-conditional blocks, each $a \in A_{\text{num}}$ and each row of D^* in P : redraw the value uniformly inside its released bin intersected with its stratification band; integer attributes draw integers (sub-bin rule)
 - (6) Return D^*
-

Selection from a pool (lines 2 to 4). The inclusion weights come from iterative proportional fitting, the raking procedure of Deming and Stephan [11], applied to the generated pool against every released cell. Exact-count batches match the release closely but not exactly, and any m rows drawn at random from a faithful pool carry the sampling error every m -record sample carries. Generation is free in ϵ , so the pipeline asks for k times the rows it needs and keeps the m whose cell counts match the release.

The pooled marginals are weighted above the per-cell counts. A cohort whose release carries only a pooled block is constrained at cohort level, not per class, because imposing a pooled histogram on each class would erase the feature-to-target structure the release spent budget on. The cost of selection is k times the generation spend.

The coverage guard (line 1). Selection can only choose among rows that exist. A pool that reached its spend cap before every cohort was complete produced a draw whose two lowest-education cohorts held 63% of the rows against a released share of 40%. Every per-row check passed; the draw simply carried the pool's imbalance. Both implementations now require every cohort to be covered by at least 1.1× the rows it owes before selecting, and the tool refuses a pool that is not.

Values below bin resolution (line 5). A released histogram fixes how many rows fall in each public bin and nothing finer, so where a value lies inside its bin is never released information. In a cohort that carries class-conditional blocks, the class shape is the release's at bin resolution, and whatever the generator does below it comes from its prior. On NHANES that prior separated the outcome classes about twice as far as the data does inside each bin: a BMI gap of 6.2 between diabetic and non-diabetic rows against 3.1 in the training data, and a positive diastolic effect the data does not have. A bin-level selection cannot see this. A tree student that splits on thresholds is indifferent to it; a linear student that fits a slope is not.

In those cohorts every numeric value is therefore redrawn uniformly inside its released bin, intersected with the row's stratification band so that no row changes cohort. Every bin count under the release's bins is unchanged by construction. In a cohort with only a pooled block, the generator's placement is the only carrier of the class signal, and it is kept; on the two pooled arms we measured, replacing it cost the tree students up to 0.03 AUC. The rule reads the release and the rows, has no parameter, and is post-processing.

4.6 Stage C: the utility transmission bound

Sections 6.3 and 6.4 measure, with access to the private data, whether the released conditional structure reaches the output. A deployment needs that measurement as a budgeted artefact a third party can inspect without that access. For each released cell c , the private data has a true positive rate p_c and the synthetic data a rate q_c . q_c is a function of the synthetic output alone, so it is public and free; p_c is private. Stage C releases a Laplace estimate $\hat{p}_c$ at a declared budget ϵ_{cert} and turns its noise into a one-sided confidence bound.

Algorithm 5: CoRTeC-Bound

Input: private D ; synthetic D^* ; the released cells $c_1, \dots, c_k$ of the finest level; suppression floor $n_{\min}$; budget ϵ_{cert} ; confidence $(1 - \alpha)$; tolerance τ

Output: a simultaneous bound B_c per cell, a verdict, or a refusal

- (1) For each cell c : $q_c \leftarrow$ positive rate of D^* in c (public)
 - (2) For each cell c : $\hat{p}_c \leftarrow p_c(D) + \text{Lap}(1/(n_{\min} \cdot \epsilon_{\text{cert}}))$ (cells are disjoint: parallel across cells)
 - (3) For each cell c : $B_c \leftarrow |\hat{p}_c - q_c| + \ln(k/\alpha) / (n_{\min} \cdot \epsilon_{\text{cert}})$
 - (4) Repeat lines 1 to 3 for a real hold-out sample in place of D^* (the ceiling) and for the same sample with its target permuted (the floor)
 - (5) If the ceiling's largest B_c exceeds τ or the floor's does not: Return "this test did not discriminate", no verdict
 - (6) Return within bound if $\max_c B_c \leq \tau$, else outside tolerance; report every B_c and the cells with thin support
-

With probability at least $(1 - \alpha)$, simultaneously over all k released cells,

$$|p_c - q_c| \leq |\hat{p}_c - q_c| + b_c \cdot \ln(k/\alpha), \quad b_c = \frac{1}{n_{\min} \cdot \epsilon_{\text{cell}}}, \quad (4)$$

where $\epsilon_{\text{cell}} = \epsilon_{\text{cert}}$ because the cells partition the data. The sensitivity of a released cell's rate is bounded by $1/n_{\min}$, the public floor, never by the private cell size, so the scale depends on nothing private. The bound is conservative by a factor $n_c / n_{\min}$ on each cell, which is the cost of using the public floor. The union bound over cells makes the guarantee simultaneous rather than per cell. The k rate queries compose in parallel, so a deployment that bounds at ϵ_{cert} has spent $\epsilon_{\text{release}} + \epsilon_{\text{cert}}$ in total.

The reference implementation reports both (`cortec.bound` in the framework, and `certify.py` in the research repository, which produced the results below). Lines 4 and 5 are the floor-and-ceiling discipline of Section 5.3 applied to the bound itself: a real sample must clear the tolerance and a permuted one must not, or the procedure issues no verdict.

The bound speaks to utility only. It says how much conditional structure survived generation, and nothing about re-identification risk. Section 7 keeps it out of the compliance mapping for that reason.

4.7 Privacy analysis

Theorem 1 (Privacy of CoRTeC). *Algorithms 1, 3 and 4 together satisfy ϵ_{total} -differential privacy. Algorithm 2 followed by them satisfies $(\epsilon_{\text{sel}} + \epsilon_{\text{release}})$ -differential privacy, which equals ϵ_{total}*

Proof. Consider neighbouring D, D' differing in one record x .

(i) *Stratification is free.* Π is a function of public attribute domains only, so $\Pi(x)$ is computed from x 's own attributes without reference to any other record. The partition structure is data-independent, and releasing it costs nothing.

(ii) *Within a cohort, sequential composition.* Fix a cohort P . Each released histogram is a counting query of L1 sensitivity 1. x contributes to exactly one bin of one attribute, so removing it changes that histogram's L1 norm by 1 and every other histogram not at all. By Proposition 1 the Laplace mechanism at scale $1/\epsilon_q$ gives ϵ_q -differential privacy per query. Under the pooled block (line 11) the q queries on P compose sequentially (Proposition 2) to $q \cdot \epsilon_q = \epsilon_{\text{marg}}$. Under the class-conditional blocks (line 9), P is partitioned by y , so x lies in exactly one of $P \cap \{y=0\}$ and $P \cap \{y=1\}$. It is touched by the class-balance query of line 7 and by the $q - 1$ histograms of its own block, and by nothing in the other block. That is a chain of q queries and the same $q \cdot \epsilon_q$. The two blocks are disjoint and compose in parallel (Proposition 3). The pooled histogram of line 10 is a function of released quantities only.

(iii) *Across cohorts, parallel composition.* The cohorts are disjoint and defined by a public rule. Record x lies in exactly one, so it can influence only that cohort's queries; the others are identical on D and D' . By Proposition 3 the cost across cohorts is ϵ_{marg} , independent of $|\Pi|$.

(iv) *Within a conditional level, parallel composition.* The cells of level ℓ partition D , so the same argument applies, and level ℓ costs ϵ_L however many cells it contains. Within a cell c we release two counting queries, the positive count k_c and the cell size $|c|$. Each has L1 sensitivity exactly 1 under add/remove adjacency, and each is noised at $\text{Lap}(2/\epsilon_L)$, a scale that depends on nothing private. They touch the same records, so they compose sequentially to ϵ_L per cell. The published rate $\rho_c = k_c / \max(\tilde{n}_c, n_{\min})$ is a function of two released quantities and the public floor, so it is post-processing (Proposition 4), as is clipping to $[0, 1]$.

(v) *Across levels, sequential composition.* Levels of differing granularity describe the same individuals, so they are not disjoint and compose sequentially: $|L_{\text{viable}}| \cdot \epsilon_L = \epsilon_{\text{cond}}$. Splitting across viable rather than declared levels is sound, because a level whose cells all fall below $n_{\min}$ releases nothing.

(vi) *The published cohort size.* $|P|$ is a counting query of sensitivity 1, noised at $\text{Lap}(1/\epsilon_{\text{count}})$. Cohorts are disjoint, so the cost across Π is ϵ_{count} by parallel composition. It composes sequentially with the q queries on the same cohort. Clamping at $n_{\min}$ is post-processing.

(vii) *Stage B is post-processing.* The row allocation (Algorithm 3, line 1) reads released sizes. The per-bin counts a batch is asked for (line 2) are apportioned from released histograms. The counts still owed (line 5) are a difference of released targets and generated rows. The prompt carries R and those counts. The model G is frozen and reads only the prompt. The acceptance rule of line 7 reads public domains. The inclusion weights, the rake, the refinement and the sub-bin redraw of Algorithm 4 read R , the generated rows and independent randomness. By Proposition 4 none of these increases the privacy loss, at any pool factor k and for arbitrarily many draws.

Summing the dependent components, $\epsilon_{\text{count}} + \epsilon_{\text{marg}} + \epsilon_{\text{cond}} = \epsilon_{\text{total}}$. Stage C, when run, adds ϵ_{cert} : its k Laplace queries are counting-rate queries of sensitivity at most $1/n_{\min}$ on disjoint cells, at a public scale, so by Proposition 3 they cost ϵ_{cert} together, and everything else in Algorithm 5 reads released or public quantities. Under auto-configuration the m_{cand} contingency tables of Algorithm 2 are counting queries of L1 sensitivity 1 on the same records and compose sequentially to ϵ_{sel} ; everything downstream of their noised counts is post-processing. The release that follows spends $\epsilon_{\text{release}}$, so the total is $\epsilon_{\text{sel}} + \epsilon_{\text{release}} = \epsilon_{\text{total}}$. ■

What the theorem does not cover. The $n_{\min}$ suppression decisions are data-dependent and uncharged. They decide which cohorts and cells appear in the output, and hence how many levels share ϵ_{cond} . The tables in this paper treat them as released, following the practice of the marginal-synthesis literature, so that our $\epsilon = 2$ rows are comparable to MST's and AIM's, whose reference implementations make the same choice. A recent audit of those implementations finds their empirical privacy close to the stated guarantee [18].

The releases behind the shipped-configuration tables (Tables 3 to 6, 9, 10, 16 and 23) and behind the class-conditional arms of Section 6.2 noise and charge their cohort counts (line 6) and their cell counts (line 13). The guarantee stated for those tables is pure $\epsilon = 2$.

The earlier-configuration arms of Section 5.5 were released by an earlier rate mechanism that noised each conditional cell's rate at a scale set from the true cell size, which step (iv) above rejects. Per conditional level that mechanism is (ϵ_L, δ) -differentially private rather than pure ϵ_L , with δ computed exactly at 1.5×10^{-7} for the $n_{\min} = 150$ floor every reported release used. For most of the project that pipeline also published per-cohort counts exact. We measured the effect of the second point on every reported number by rebuilding each affected draw under noised counts: the largest change on any metric was 0.0005, below draw-to-draw variation. The headline Adult arm regenerated under the corrected two-count mechanism reproduces the earlier result (Section 6.1).

The floating-point attack of Mironov [31] on the Laplace mechanism applies to every implementation in the comparison, including ours. It is declared as an open gap in the release's own audit block.

4.8 What the allocation achieves

Against a natural implementation (private clustering for cohorts, basic composition across cohorts, and separate mean and standard-deviation queries), the same ϵ_{total} yields an ϵ per released statistic of 0.0667 against 0.0100, a 6.7 $\times$ improvement. The realised per-query budget on the 15-column Adult schema at $\alpha = 0.5$ is 0.0653, because the cohort-size query of line 6 takes its 2% first; on the same footing the improvement is 6.5 $\times$.

The budget is verified by instrumentation rather than by re-deriving a model of it. A test patches the Laplace mechanism inside the release path and tallies what is actually spent, asserting exactly ϵ_{total} on three auto-configured schemas and at three budgets.

A tighter accountant would not help. Advanced composition [15] and concentrated differential privacy [6] loosen the bound by 16 to 31% at the chain lengths this method reaches, and only cross over past 27 queries. Those chain lengths are 15 marginal queries per cohort on the Adult schema, or 18 once its three conditional levels are counted. No schema we tested has a conditional hierarchy deeper than 3 levels, or deeper than 2 at the shipped $n_{\min}$.

4.9 Using the mechanism

Figure 2 shows the complete workflow in the reference implementation. Stage A is the only line that touches private data. The release is serialised with its audit trail and contains no record. Stage B may be run as often as wanted, on any endpoint, at no further privacy cost.

```

1  from cortec import Schema, Band, release_statistics, Generator
2
3  schema = Schema(
4      name="encounters",
5      numerical={"age": (18, 95), "length_of_stay": (1, 30)},
6      categorical={"admission_type": ["Emergency", "Elective", "Urgent"],
7                  "alc_result": ["None", "Norm", ">7", ">8"]},
8      target="readmitted_30d", positive="YES", negative="NO",
9      bins={"age": [18, 40, 55, 70, 95], "length_of_stay": [1, 3, 6, 10, 30]},
10     stratify=[("length_of_stay", [Band("los1-2", 1, 3), Band("los3-5", 3, 6),
11                                     Band("los6+", 6, 31)])],
12     conditional=[("admission_type",), ("admission_type", "alc_result")],
13 )
14
15 release = release_statistics(schema, private_df, epsilon_total=2.0, n_min=150)
16 print(release.audit["epsilon_accounted"])
17 release.to_json("release.json")
18
19 gen = Generator(schema, backend="gemini", surface="vertex",
20                 model="gemini-3.5-flash", reasoning="on")
21 synthetic = gen.generate_selected(release, n_rows=5000, pool_factor=3)
22 print(gen.stats.calls_with_reasoning_block, "of", gen.stats.calls, "calls reasoned")

```

Figure 2: The reference implementation end to end. The schema declares only what an institution knows without looking at its data: column names, public bounds and bin edges, the target, and the cohort and conditional structure. `cortec.autoconfig.derive()` can derive the last of these instead. The call to `release_statistics` is the only line that touches private data; it prints an accounted budget of 2.0 and writes a release that contains no private record. `generate_selected` runs Stage B with the shipped defaults: exact-count batches, a 3 $\times$ pool, selection to the release and the sub-bin rule. The final line reports on how many calls the model's reasoning verifiably fired.

5. Experimental Setup

5.1 Datasets

Table 2: Datasets. The three in bold carry the head-to-head of Section 6.1.

dataset	domain	records	attributes	positive rate	rows per person	role
NHANES 2017–2018 [9]	healthcare (examination and laboratory)	3,749	10	14.2%	1	primary clinical dataset
UCI Adult [12, 25]	census	32,561	15	24.1%	1	comparability anchor; AIM’s own primary dataset
Default of credit card clients [51]	finance	30,000	15	22.1%	1	second regulated domain
Diabetes 130-US hospitals [42]	healthcare (administrative)	101,763	19	11.2%	mean 1.42, max 40	ϵ sweep at scale; transmission; membership inference
renal registry (constructed)	synthetic clinical	20,000	10	14.9%	1	contamination control (Section 6.3)
six untouched benchmarks	3 health, 3 finance	303–45,211	8–20	6.0–53.1%	1	auto-configuration generalisation (Section 6.7)

NHANES 2017–2018 is the primary clinical dataset: a national health examination survey of physical measurements and laboratory assays, obtainable with no application. We join the demographic, glycohaemoglobin, body-measures and blood-pressure files on the respondent identifier and restrict to adults aged 18 to 80. The target is $\text{HbA1c} \geq 6.5\%$, the American Diabetes Association’s diagnostic threshold [4], which makes this an undiagnosed-diabetes screening task on laboratory values rather than on billing codes. Bin edges are public clinical knowledge (WHO BMI categories, ACC/AHA blood-pressure stages), not quantiles of the file.

Adult is the head-to-head anchor, the only schema on which every mechanism we compare completes. The credit dataset is the second regulated domain. Diabetes 130 is hospital administrative data at scale, with 81,410 training records; it carries the ϵ sweep on a large dataset, the transmission sweep and part of the membership-inference evaluation. It is encounter-level, so every figure from it holds under a row-level guarantee (Section 3).

Each dataset is split once into a training set, which every mechanism sees, and a held-out test set, which only the evaluator sees. There are 26,048 training rows on Adult, 24,000 on finance and 2,999 on NHANES.

5.2 Baselines

We compare against the three families an institution would realistically choose between: the marginal methods MST and AIM; the DP-SGD generative model DP-CTGAN; and the PATE generative models PATE-CTGAN and PATE-GAN. All run through smartnoise-synth [34] at matched ϵ . Numerical columns are pre-binned using the same public bounds CoRTeC assumes, so no method gains an advantage from a different discretisation. All run at library defaults at $\epsilon = 2$ on the training split and are sampled at the same n as CoRTeC.

An unconditioned “header-only” prompt, the same frozen model given nothing but the column names, appears as a control and not as a competitor. It carries no privacy guarantee. Its role is to establish that CoRTeC’s output depends on the released statistics rather than on the model’s prior.

5.3 Metrics, and the floor-and-ceiling discipline

We report:

- marginal fidelity: 1-way and 2-way total variation, meaning half the summed absolute difference between two histograms, the latter over all column pairs;
- conditional fidelity: error in $P(\text{target} \mid \text{group})$ over the groups the release describes (“seen”) and over strictly held-out group families sharing no column with the conditioning (“held-out”);
- AIM’s own 3-way workload error where stated; and
- utility: TSTR AUC under logistic regression, random forest and gradient boosting, each trained on the synthetic data and tested on the real held-out split.

Fidelity and utility appear in the same table for every condition, because a method can score well on one and poorly on the other, and both marginal baselines do.

Two references accompany every table. The *real-sample floor* is a genuine draw of n real training records. No synthetic method can be expected to exceed it at that n , and it is the only sensible target. The *permuted-target floor* is real data with the target column permuted. It has identical marginals and zero predictive content, and is therefore the score of data that carries no usable information. Any metric that fails to separate these two has no power, and we check that before drawing conclusions (Section 6.3). The *train-on-real ceiling* trains the students on the entire training set.

Evaluating against explicit reference points rather than a bare threshold is the discipline DPBench argues for [19], and it caught the founding defect of this project. Our original primary metric scored the real-data ceiling and the unconditioned baseline identically, so there was nothing for any method to improve, and a verdict had been drawn from it before the floors were checked (Section 9). The real-sample floor is drawn afresh for each table at that table's n and draw count, so its value varies across tables by design. Comparisons are made at matched n .

Every metric is computed by one implementation that bins numerical columns with the public bin edges every method is fed. One convention is declared. The evaluator's intervals are right-closed, while the release, the selection step and the baselines' discretisation are left-closed, so an integer that lies exactly on an edge counts in different bins under the two. The effect is small and falls on every method. On Adult, 1-way error under the release's convention is 0.020 for CoRTeC, 0.015 for MST, 0.030 for AIM and 0.043 for a real sample, against 0.029, 0.024, 0.037 and 0.043 under the evaluator's. Every number in this paper is the evaluator's, and the ordering of methods is the same under either.

5.4 Statistics

Point estimates are means over draws. Differences between conditions are tested with Welch's t-test on the per-draw values and corrected with Holm's procedure [20] over each family of tests, a family being the seven metrics of one baseline on one dataset. Adjusted p-values are Holm's step-down values. Where a rank test is reported, its smallest attainable p-value at the draw counts in hand is reported beside it: at small group sizes it is the design, not the data, that limits the statistic. Confidence intervals on the headline equivalence test are bootstrap intervals over draws (10,000 resamples).

All CoRTeC draws within a condition come from one release and one generator, so they replicate the generation step. Releases are redrawn only for the transmission sweeps. Draws from different generators are never pooled, because the generator is itself a variable under study (Section 6.6).

5.5 The configuration reported

Unless a table says otherwise, every CoRTeC row is the shipped configuration:

- the class-conditional release of Algorithm 1 at $\epsilon = 2$ with $\alpha = 0.2$ and $n_{\min} = 150$;
- exact-count batches of 25 rows;
- a $3\times$ pool ($2\times$ on NHANES) selected to the release with the sub-bin rule;
- Gemini 3.5 Flash with reasoning enabled, reached through Vertex AI inside a Google Cloud project; and
- $n = 300$ and three draws per dataset.

Adult and finance use hand-tuned cohort and conditional hierarchies. NHANES uses the auto-configured release of Algorithm 2: three age cohorts and two conditional levels over age, diastolic blood pressure and race/ethnicity.

Several results were produced on the earlier configuration the method was developed on, and each is labelled where it appears. They are:

- the powered six-draw equivalence test of Section 6.1;
- the 3-way workload table (Table 7);
- the saturation and inversion controls of Section 6.3;
- the transmission sweeps and the over-coupling decomposition of Section 6.4;
- the Diabetes 130 budget sweep of Section 6.5;
- the generator ladder of Section 6.6;
- the cohort-wise and cell-wise comparison of Section 6.8; and

- the membership-inference evaluation of Section 6.11.

That configuration differed from the shipped one in four ways. It released pooled rather than class-conditional histograms. It stated the release to the generator as shares rather than exact counts. It kept every returned row rather than selecting from a pool. And, in the six-draw Adult arm, it showed the generator only the four most common values of each categorical attribute. Every one of those differences is conservative for CoRTeC in a comparison. The conclusions those sections draw concern the variable each isolates, not the configuration.

6. Results

6.1 Head-to-head: fidelity at the release’s own level, utility at the floor

The bound first. No output decoded from a release can be expected to be closer to the private data than the release itself. At $\epsilon = 2$ with a fifth of the budget on the conditional table, the release’s 1-way total variation against the training data is 0.020 on Adult and 0.013 on finance under the public bins. A 300-record real sample records 0.043 and 0.041 on the same measure. The release therefore carries less marginal error than a sample of the size we generate.

Adult. Table 3 reports three draws from one release beside the baselines at five draws each. CoRTeC’s 1-way error, 0.029, is below the real sample’s 0.043 and within 0.005 of MST’s 0.024; that difference survives correction (adjusted $p = 0.007$) and is the one comparison in this section in which a baseline’s error is lower. Its 2-way error, 0.110, equals the real sample’s 0.107 within noise and is below AIM’s 0.119 and MST’s 0.144. Conditional error over seen groups, 0.017, is a quarter of the real sample’s, because the class-conditional blocks are estimated from all 26,048 training records and a 300-record sample is not. Over held-out groups it is 0.039 against the sample’s 0.036. The three students train at 0.842, 0.876 and 0.849 against the sample’s 0.830, 0.870 and 0.840. The difference from AIM and MST on every student is 0.14 to 0.17 AUC.

Table 3: Adult at $\epsilon = 2$, $n = 300$.

condition	draws	1-way TV ↓	2-way TV ↓	cond. seen ↓	cond. held-out ↓	TSTR-LR ↑	TSTR-RF ↑	TSTR-GBM ↑
<i>train on all 26,048 real rows (ceiling)</i>	1	0.011	0.028	0.020	0.008	0.860	0.907	0.914
<i>real sample, $n = 300$ (floor)</i>	3	0.043	0.107	0.062	0.036	0.830	0.870	0.840
CoRTeC (Gemini 3.5 Flash)	3	0.029	0.110	0.017	0.039	0.842	0.876	0.849
CoRTeC (Claude Opus 5)	1	0.030	0.110	0.013	0.042	0.838	0.870	0.825
MST	5	0.024	0.144	0.169	0.050	0.690	0.728	0.681
AIM	5	0.037	0.119	0.054	0.101	0.675	0.700	0.691
<i>permuted target (no-information floor)</i>	3	0.041	0.110	0.152	0.102	0.423	0.485	0.482

Finance. Table 4 reports three draws from one release beside MST and PATE-CTGAN on the full 15-column schema, and beside AIM on the reduced 12-column schema. That schema is the only one AIM completes on within three hours; its fit on the full schema exceeded three hours on every attempt, and the cause is localised to the six high-cardinality amount columns. CoRTeC, MST and the floors are re-scored on those 12 columns for that comparison.

The finance result has the same shape as the Adult result. 1-way error, 0.029, is within 0.001 of MST’s 0.028 (the two do not separate) and below the real sample’s. 2-way error, 0.125, is below MST’s 0.230 and above the sample’s 0.099; on the 12-column schema it is 0.107 against AIM’s 0.136. Conditional error is below the sample’s on seen and held-out groups. The tree students match the real-sample floor (0.731 against 0.727, 0.712 against 0.717). Logistic regression is 0.015 short (0.680 against 0.695, $p = 0.033$ at three draws against three). On the 12-column schema CoRTeC’s 1-way error is lower than AIM’s by a factor of 2.1, and its students are higher on every one. It does not separate from MST on 1-way error, the measure MST is built for.

Table 4: Finance at $\epsilon = 2$, $n = 300$. AIM completes only on the 12-column schema, so that block re-scores every arm on those columns.

condition	draws	1-way TV ↓	2-way TV ↓	cond. seen ↓	cond. held-out ↓	TSTR-LR ↑	TSTR-RF ↑	TSTR-GBM ↑
<i>train on all 24,000 real rows (ceiling)</i>	1	0.009	0.024	0.013	0.008	0.699	0.748	0.747
<i>real sample, n = 300 (floor)</i>	3	0.041	0.099	0.059	0.045	0.695	0.727	0.717
CoRTeC (Gemini 3.5 Flash)	3	0.029	0.125	0.014	0.022	0.680	0.731	0.712
MST	3	0.028	0.230	0.090	0.150	0.580	0.610	0.604
PATE-CTGAN	3	0.209	0.397	0.192	0.138	0.548	0.501	0.515
<i>permuted target (no-information floor)</i>	3	0.040	0.100	0.137	0.059	0.514	0.482	0.481
12-column schema: CoRTeC	3	0.026	0.107	0.014	0.022	0.678	0.729	0.704
12-column schema: AIM	5	0.054	0.136	0.083	0.036	0.578	0.652	0.623
12-column schema: MST	3	0.023	0.216	0.090	0.150	0.591	0.644	0.636
<i>12-column schema: real sample, n = 300</i>	3	0.038	0.090	0.059	0.045	0.682	0.722	0.709

NHANES, auto-configured. Table 5 is the turn-key case: a release the auto-configurator derived with no hand-tuned hierarchy, generated cohort-wise from a $2\times$ pool, beside MST, AIM and PATE-CTGAN fitted on NHANES at the same ϵ and binning. NHANES is also the one clinical dataset on which AIM completes (58 s to fit), so it carries the healthcare panel of Figure 3.

The release’s 1-way error against the training data is 0.009, and the selected draws’ 1-way error against the release is 0.002. The marginal result therefore carries over to a release nobody tuned: 1-way error 0.015 against the sample’s 0.034, conditional error over seen groups 0.006 against 0.015, held-out 0.036 against 0.036.

Two things do not carry over, and both are stated. 2-way error is 0.119 against the sample’s 0.081 (Welch $p = 0.001$): with three cohorts and a four-cell table the release names little pairwise structure, and what the selection does not constrain the generator supplies. And logistic regression trains at 0.752 against 0.773 ($p = 0.047$), while the tree students match the floor (0.732 against 0.728, 0.711 against 0.716). The remaining linear gap is not in the release, since the release decoded with no model at all trains logistic regression at 0.767. The gap lies in the youngest cohort, whose diabetic class does not clear $n_{\min}$; that cohort therefore keeps a pooled block and the generator’s values below bin resolution (Section 4.5). CoRTeC’s error is lower than all three baselines’ on every measure except conditional error over seen groups, where MST’s 0.004 and CoRTeC’s 0.006 do not separate.

Table 5: NHANES at $\epsilon = 2$, $n = 300$, under a release nobody tuned.

condition	draws	1-way TV ↓	2-way TV ↓	cond. seen ↓	cond. held-out ↓	TSTR-LR ↑	TSTR-RF ↑	TSTR-GBM ↑
<i>train on all 2,999 real rows (ceiling)</i>	1					0.797	0.763	0.747
<i>real sample, n = 300 (floor)</i>	3	0.034	0.081	0.015	0.036	0.773	0.728	0.716
CoRTeC (Gemini 3.5 Flash, auto-configured)	3	0.015	0.119	0.006	0.036	0.752	0.732	0.711
MST	3	0.024	0.175	0.004	0.128	0.687	0.646	0.664
AIM	3	0.024	0.142	0.009	0.073	0.596	0.650	0.650
PATE-CTGAN	3	0.095	0.187	0.149	0.150	0.598	0.552	0.539
<i>permuted target (no-information floor)</i>	3	0.043	0.093	0.047	0.060	0.493	0.514	0.515

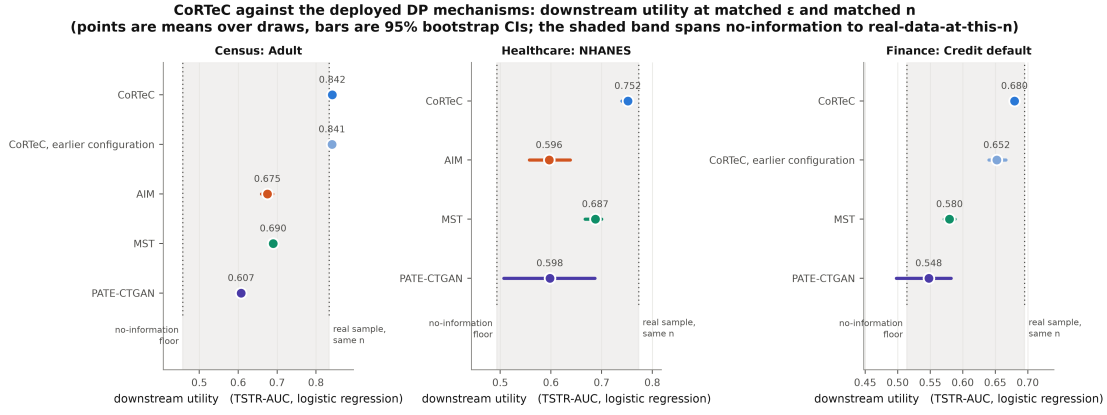


Figure 3: The central comparison on three datasets. Points are means over draws, bars are 95% bootstrap confidence intervals, and the shaded band spans the no-information floor to a real sample of the same size.

The family statistics. Table 6 gives the Welch tests on the per-draw values, Holm-corrected over each seven-metric family of one baseline on one dataset, in the direction that favours CoRTeC. On Adult and finance, 28 of the 35 comparisons survive correction in CoRTeC’s favour and one separates in a baseline’s favour: MST’s 0.005 advantage on 1-way error on Adult. The six that do not separate are:

- the two 1-way comparisons against MST on finance, where the two methods are within 0.003;
- conditional error over seen groups against MST on finance, on both schemas;
- TSTR-GBM against MST on the full finance schema; and
- held-out conditional error against AIM on the 12-column schema.

On NHANES, 8 of the 21 comparisons survive at three draws against three. Six of them are fidelity measures, and two are the random-forest student against MST and AIM (+0.086 and +0.083). The other utility differences on NHANES are consistent in direction and do not survive at three draws. The utility ordering there is therefore a corrected result on one student and a signal on the other two.

Table 6: Holm-corrected families. Cells are the difference CoRTeC – baseline with the adjusted p in brackets. Negative error differences and positive AUC differences favour CoRTeC.

family	1-way	2-way	cond. seen	cond. held-out	TSTR-LR	TSTR-RF	TSTR-GBM
Adult, against AIM	−0.007 (0.0029)	−0.009 (0.0029)	−0.037 (0.0003)	−0.061 (0.0016)	+0.167 (<0.0001)	+0.176 (0.0020)	+0.157 (0.0002)
Adult, against MST	+0.005 (0.0074)	−0.034 (0.0001)	−0.152 (0.0008)	−0.011 (0.0015)	+0.152 (0.0008)	+0.148 (<0.0001)	+0.167 (0.0004)
finance, against MST	+0.001 (0.14)	−0.105 (0.0074)	−0.076 (0.063)	−0.128 (0.0005)	+0.100 (0.0024)	+0.122 (0.0074)	+0.108 (0.051)
finance, 12 columns, against AIM	−0.028 (0.0023)	−0.029 (0.022)	−0.068 (0.0023)	−0.014 (0.17)	+0.100 (0.022)	+0.077 (0.010)	+0.082 (0.010)
finance, 12 columns, against MST	+0.003 (0.082)	−0.109 (0.010)	−0.076 (0.082)	−0.128 (0.0009)	+0.087 (0.0009)	+0.085 (0.010)	+0.069 (0.026)
NHANES, against MST	−0.009 (0.045)	−0.056 (0.0053)	+0.002 (0.41)	−0.092 (0.045)	+0.064 (0.090)	+0.086 (0.033)	+0.046 (0.41)
NHANES, against AIM	−0.009 (0.41)	−0.023 (0.078)	−0.003 (0.41)	−0.037 (0.0083)	+0.155 (0.14)	+0.083 (0.027)	+0.060 (0.24)
NHANES, against PATE-CTGAN	−0.080 (0.043)	−0.068 (0.0083)	−0.143 (0.078)	−0.114 (0.078)	+0.154 (0.41)	+0.181 (0.17)	+0.171 (0.065)

The powered equivalence test. The claim that a model trained on CoRTeC’s output performs as one trained on a real sample of the same size was first tested with adequate power on Adult. The generator was Claude Fable 5, with six draws against five real samples, on the earlier release design; the prompt showed only the four most common values of each categorical attribute, the weaker of the two prompts we ran. The differences were +0.007, −0.003 and +0.012 AUC on the three students. The 95% confidence intervals on the difference were [−0.007, +0.021], [−0.015, +0.010] and [−0.000, +0.027], and every $p > 0.18$.

A non-significant difference is not by itself evidence of equivalence. What makes this one an equivalence result is that the intervals exclude any advantage or deficit larger than about 0.027 AUC. The same arm regenerated from a fresh release under the two-count rate mechanism of Algorithm 1 reproduces it (0.848, 0.871 and 0.852 against 0.834, 0.872 and 0.843; the one student that moves, moves in CoRTeC’s favour, $p = 0.050$). Tables 3 to 5 remake the result under the shipped configuration and a second generator family on three datasets, at three draws each.

A second vendor. The Adult release was also generated through Claude Opus 5, one complete threefold pool of 900 rows selected exactly as the Gemini pools were. The pool’s 1-way error against the released histograms is 0.007, lower than the Gemini pools’ 0.009 to 0.011. The selected draw reproduces the result (Table 3): each measure is within 0.024 of the Gemini means, and the three students match the real-sample floor. One draw is a check rather than a replication, and we present it as one.

Reading the numbers below the real-sample floor. CoRTeC’s 1-way and conditional errors are below what a 300-record real sample achieves, as MST’s 1-way error is. An output selected to match statistics estimated from the whole training set is smoother than a random sample of the same size, not more faithful to one. The comparison that settles whether that smoothness costs anything is the downstream one, and it does not: the students match the real-sample floor. Where the deliverable is a dataset meant to stand in for a random sample of a given size, the sampling variability a real sample carries is part of what is being imitated, and a selected output does not carry it.

The generative families. All three fail on Adult at $\epsilon = 2$, and they fail by not preserving the target’s base rate, in opposite directions. DP-CTGAN collapses towards the negative class: two of its three draws contain no positive record at all, so TSTR is undefined rather than poor, and the fit took 2.2 hours. PATE-GAN and PATE-CTGAN fail the other way, emitting 54 to 60% and 45% positives against a real 24.1% (TSTR-LR 0.562 and 0.608). PATE-GAN’s conditional error over seen groups (0.525) is worse than the permuted-target floor (0.155). These are library defaults at this budget on a 15-attribute schema, and we make no claim that they cannot be made to work with tuning. What we can say is that the failure is invisible in a marginal-fidelity summary unless the base rate is inspected.

AIM’s own workloads. Table 7 reports the metric AIM is built for, at a sample size adequate for 3-way marginals. AIM records the lowest error on all three of its workloads, CoRTeC the second lowest, and MST and PATE-CTGAN follow. At $n = 300$ the same comparison gives CoRTeC lower error than AIM on the target workload; that ordering inverts at adequate n because the small-sample floor masks AIM’s advantage, so high-order marginal comparisons at small n are not trustworthy. AIM’s advantage on its own workloads is structural rather than a shortfall in CoRTeC. Its adaptive measurement selection optimises exactly that workload under the budget, so the lowest error there is what its design predicts, and we did not set out to match it.

Table 7: 3-way workload error on Adult at $n = 1,950$ per method (three draws each; the CoRTeC arm is the earlier configuration).

method	1-way TV ↓	2-way TV ↓	target-3way ↓	all-3way ↓	skewed-3way ↓
<i>real sample (floor at n)</i>	<i>0.017</i>	<i>0.042</i>	<i>0.120</i>	<i>0.179</i>	<i>0.166</i>
AIM	0.024	0.076	0.278	0.279	0.252
MST	0.024	0.130	0.444	0.543	0.491
CoRTeC (Claude Fable 5)	0.054	0.116	0.305	0.389	0.357
PATE-CTGAN	0.357	0.538	1.190	1.363	1.336

6.2 What the release carries, and what each step of Stage B contributes

The release alone, decoded with no model. The clearest way to see what a release carries is to decode it by independent sampling, with no language model and no prior. A cohort is drawn by its released size, the outcome from the released class balance, and every other column from that outcome’s histogram block. Under a pooled release, the columns come from the cohort’s single block and the outcome from the conditional table. Table 8 does this five times for each release design on finance and on Adult.

Decoded this way, the class-conditional release trains better models than the pooled one on every student and both datasets. The gain is 0.02 to 0.04 AUC on finance (Welch $p = 0.089, 0.048, 0.013$ for LR, RF, GBM) and 0.05 to 0.06 on Adult ($p \leq 0.005$ on all three). The class-conditional release is within 0.011 AUC of the real-sample floor on every finance student and within 0.019 on every Adult student. Held-out conditional error falls by 31 to 39%. The pooled release, decoded the same way, gives the same utility as our earliest generated finance arm did: the

information a classifier needs was not in the release the generator was given. The naive decode is not a competitor to the generator, since its 2-way error is 1.6× the real sample’s. It is a lower bound on the information the release contains.

Table 8: Each release design decoded by independent sampling, with no model (five decodes each).

release, decoded naively	1-way TV ↓	2-way TV ↓	cond. seen ↓	cond. held-out ↓	TSTR-LR ↑	TSTR-RF ↑	TSTR-GBM ↑
finance , pooled	0.053	0.157	0.053	0.042	0.666	0.701	0.664
finance , class-conditional	0.054	0.161	0.061	0.029	0.687	0.733	0.706
<i>finance</i> , real sample $n = 300$	<i>0.041</i>	<i>0.099</i>	<i>0.059</i>	<i>0.045</i>	<i>0.695</i>	<i>0.727</i>	<i>0.717</i>
Adult , pooled	0.052	0.144	0.069	0.094	0.752	0.814	0.792
Adult , class-conditional	0.058	0.150	0.070	0.057	0.811	0.861	0.842
<i>Adult</i> , real sample $n = 300$	<i>0.043</i>	<i>0.107</i>	<i>0.062</i>	<i>0.036</i>	<i>0.830</i>	<i>0.870</i>	<i>0.840</i>

The same generator on both releases confirms it. With one finance release of each kind, Gemini 3.5 Flash, the same prompts and three draws per release, the class-conditional release raises every student: +0.029, +0.050 and +0.042 AUC (Welch $p = 0.007, 0.017, 0.029$; Hedges’ g , a standardised effect size, of 2.4 to 4.0). The four fidelity measures stay within draw-to-draw spread ($p \geq 0.20$). The pooled arm under Gemini reproduces the pooled arm under Claude Table 5 to within 0.012 AUC on every student (0.651, 0.662, 0.662 against 0.652, 0.673, 0.664), which is direct confirmation that the ceiling was the release’s, not the generator’s.

Under Claude Table 5 on the same release design the tree students again reach the floor (0.723 and 0.716 against 0.727 and 0.717), but held-out conditional error rises from 0.048 to 0.057 ($p = 0.007$), a cost the Gemini arm does not show. The per-outcome blocks give that generator more to honour per cohort, and the cost appears on the conditional families the release does not name. We report it as a cost of the design under one generator rather than as noise.

Exact counts and selection. The two Stage B steps were separated on the Adult release by scoring the same generated rows before and after selection (Table 9). Exact counts act on the pool: the three generated pools (879, 900 and 900 rows) match the released histograms to 0.009 to 0.011, below the release’s own distance from the truth, whereas the plain share-based prompt’s rows have an error of 0.052 against theirs. But any 300 rows drawn at random from a faithful pool carry the sampling error every 300-record sample carries, and a random 300 rows from that pool have a 1-way error of 0.050, close to the real sample’s 0.043.

Selection removes that sampling error. 1-way error falls from 0.050 to 0.029, 2-way error equals the real sample’s, conditional error falls by almost three quarters, and the three students move up by 0.009 to 0.027 AUC. The two steps are a pair. Counts without selection give a pool the selection can trust; selection without counts chooses from rows that were never asked to match, which produced no gain that survived a significance test on the earlier arms.

Table 9: Exact counts and selection, separated on one Adult release with three pools. The selected row is re-selected from the same three pools at a fixed seed, which is why its 1-way error coincides with Table 3’s while the students differ slightly.

condition	1-way TV ↓	2-way TV ↓	cond. seen ↓	cond. held-out ↓	TSTR-LR ↑	TSTR-RF ↑	TSTR-GBM ↑
exact-count batches, a random 300 of each pool	0.050	0.131	0.057	0.052	0.826	0.851	0.821
the same pools, the 300 rows selected to the release	0.029	0.109	0.015	0.037	0.835	0.872	0.848
<i>real sample</i> , $n = 300$	<i>0.043</i>	<i>0.107</i>	<i>0.062</i>	<i>0.036</i>	<i>0.830</i>	<i>0.870</i>	<i>0.840</i>
<i>the release itself against the training data</i>	<i>0.020</i>						

The sub-bin rule. Table 10 applies the rule to every stored arm, with the values the generator produced beside the values the rule produces. It does what it was built for on the arm that needed it, and nothing on the arms that did not. On the class-conditional arms of Adult and finance the students move by at most 0.010 AUC, in both directions, inside draw-to-draw spread. The pooled-release arms are untouched by construction.

Two alternatives were measured on the same arms and rejected because they did not generalise. A redraw applied to every cohort regardless of blocks raised the NHANES linear student further, to 0.765, but it cost the tree students 0.03 on NHANES at $n = 600$ and 0.016 on Diabetes 130, where the pooled release leaves the generator’s placement as the only class signal. A calibration of each block’s mean to the value its released histogram implies moved rows across stratification bands, and was abandoned.

Table 10: The sub-bin rule on every stored arm: the generator’s values, then the release’s.

arm	draws, n	TSTR-LR	TSTR-RF	TSTR-GBM	1-way TV	real sample LR / RF / GBM
NHANES, shipped configuration	3, 300	0.729 to 0.752	0.728 to 0.732	0.712 to 0.711	0.018 to 0.015	0.773 / 0.728 / 0.716
Adult, shipped configuration (Gemini)	3, 300	0.842 to 0.842	0.872 to 0.876	0.842 to 0.849	0.026 to 0.029	0.830 / 0.870 / 0.840
Adult, shipped configuration (Opus 5)	1, 300	0.840 to 0.838	0.872 to 0.870	0.833 to 0.825	0.027 to 0.030	0.830 / 0.870 / 0.840
finance, shipped configuration (Gemini)	3, 300	0.684 to 0.679	0.721 to 0.731	0.707 to 0.712	0.028 to 0.029	0.695 / 0.727 / 0.717
NHANES, pooled release	1, 600	0.763 to 0.763	0.744 to 0.744	0.733 to 0.733	0.025 to 0.025	0.778 / 0.735 / 0.725
Diabetes 130, pooled release	1, 300	0.555 to 0.555	0.570 to 0.570	0.566 to 0.566	0.053 to 0.053	0.557 / 0.556 / 0.543

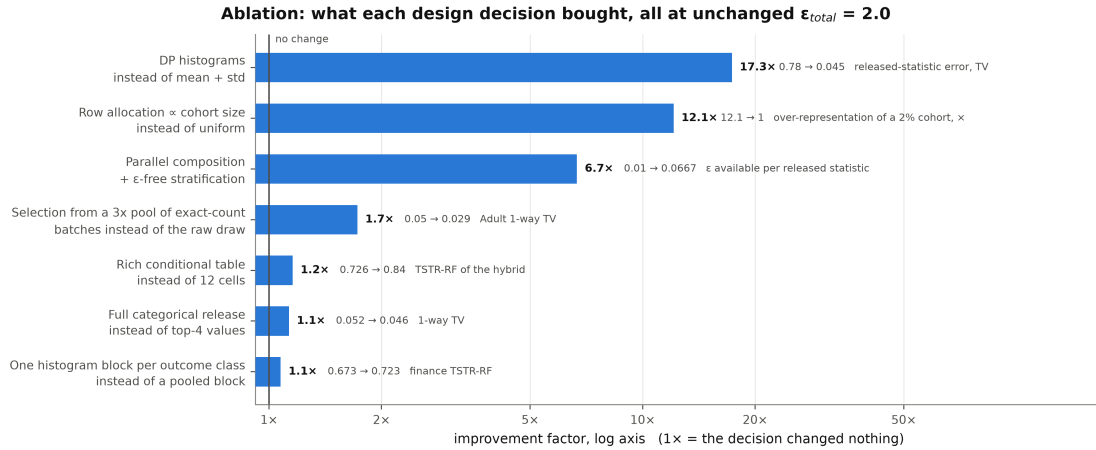


Figure 4: What each Stage A design decision changed, each on the axis it targets. Parallel composition with free stratification: ϵ per statistic 0.0100 to 0.0667. Histograms in place of mean and standard deviation: released-statistic error 0.78 to 0.045 total variation. Row allocation by released mass: a 2%-of-population cohort from 12.1x to 1.0x over-represented. Showing the generator the full released categorical distributions rather than the four most common values: 1-way error 0.052 to 0.045, TSTR-LR 0.841 to 0.851.

6.3 Both standard acceptance criteria are saturated

Marginal fidelity. A common statement of the fidelity goal is that synthetic data should be “at least 90% statistically similar” to the private data, operationalised as $1 - TV$. Table 11 tests whether that criterion can tell good data from useless data, using the permuted-target control on Adult. The permuted control has no usable feature-to-target relationship (TSTR 0.458, near chance) and yet records 95.7% 1-way similarity, passing a 90% bar, because permuting one column of fifteen leaves every marginal intact. Tightening the bar far enough to exclude the useless case also excludes a genuine sample: on 2-way similarity the pair is 0.106 and 0.112, so a 90% two-way bar rejects both, including real data drawn from the private distribution itself. A criterion that fails real data is not a usable acceptance test.

Table 11: A marginal criterion cannot separate good data from useless data (Adult, $n = 300$).

metric	real sample (known good)	permuted target (known bad)	separates?
1-way TV	0.042	0.043	no; saturated
2-way TV	0.106	0.112	no; saturated
conditional TV (seen)	0.063	0.155	yes
conditional TV (held-out)	0.036	0.102	yes

metric	real sample (known good)	permuted target (known bad)	separates?
TSTR-LR	0.834	0.458	yes

We therefore propose that any fidelity criterion for private synthesis carry a conditional term. A concrete instance that real data and CoRTeC pass, while AIM, MST, PATE-CTGAN, DP-CTGAN, PATE-GAN and the permuted floor all fail, is: 1-way TV ≤ 0.06 , 2-way TV ≤ 0.135 , and mean conditional TV ≤ 0.07 . The separating work is done by the conditional term. AIM fails it at 0.077, MST at 0.110 and the permuted floor at 0.129, while every method that passes is near 0.05. The marginal thresholds exclude the deep generative baselines, which miss by an order of magnitude, and MST, whose 2-way error of 0.144 is above the bar. The 2-way threshold depends on how pairs are counted: it was 0.12 under a 20-pair sample and is 0.135 over all 105 pairs, for the same data. The conditional term’s scale did not move, which is one more reason to let it carry the criterion.

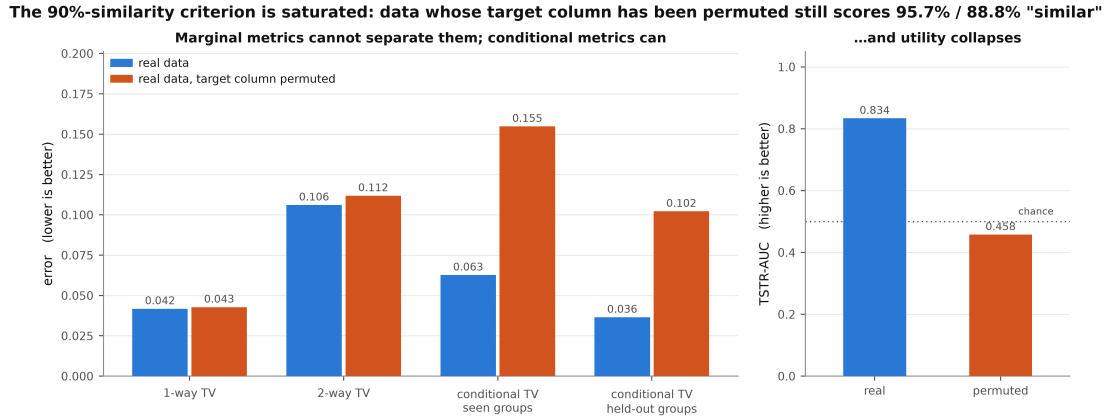


Figure 5: Real data against the same data with its target column permuted. Marginal metrics cannot separate them; conditional metrics and utility can.

Aggregate utility. It does not follow that aggregate utility is safe. The header-only control gives the same frozen model nothing but the column names. On Adult it reaches TSTR-LR 0.824, against a real 300-record sample’s 0.834, while AIM reaches 0.675 and MST 0.690. On that dataset, then, an ungrounded language model with no access to the private data at all comes within 0.010 of a real sample and exceeds AIM and MST by 0.13 to 0.15 on the metric the field uses to argue for downstream usefulness. That is a result about the metric and about the datasets: on public benchmarks a pretrained model already encodes most of the aggregate structure, so aggregate utility cannot distinguish a method that reads the private release from one that recites a prior.

Making the prior wrong. We can separate the two by inverting a relationship the prior knows. In the inversion condition the released statistics state that advanced-degree holders (`education_num` ≥ 14) earn over 50K at a rate of 0.2%, against a real-world rate of 61.9%. The control is matched: it takes CoRTeC’s own prompt and strips exactly the released blocks (class balance, histograms, categorical proportions, conditional table, cohort size), leaving every other line byte-identical, and it runs through the same generation routine, so row allocation, batching, parsing and repair are shared by construction.

With one release, Claude Fable 5 and $n = 300$ per condition, CoRTeC emits the high-income label for 0.000 of advanced-degree holders, tracking the release. The matched control emits it for 0.931, more extreme than the real-world prior (Table 12). Since every other element of the two conditions is identical, the separation is attributable to the released statistics alone. The control is told to match a table and to ignore world knowledge; it has no table, and answers from its prior. We report one draw per arm because the measure spans the full $[0, 1]$ range: draw-to-draw variation on conditional quantities in this paper is 0.004 and release-to-release variation is 0.037, two orders of magnitude below the effect.

Table 12: The inversion test. The two conditions are identical but for the released arrays.

matched condition	emits the high-income label for advanced degrees
CoRTeC	0.000 , tracking the release
matched header-only control	0.931 , more extreme than the real-world prior
real data, uninverted	0.619

Evaluated on a held-out test set carrying the same inverted relationship, the model trained on header-only output is wrong for 72% of the inverted subgroup. Its aggregate AUC still exceeds CoRTeC’s (0.730 against 0.670), because the subgroup is 7.9% of rows and the aggregate is dominated by the 92% where the prior is correct.

On the uninverted release the same matched control separates too, at two independent replications of the whole pipeline. CoRTeC’s TSTR-LR is 0.057 higher than the control’s (0.849 against 0.792) against a pooled draw-to-draw spread of 0.001, and its conditional error is 3.7× lower (0.043 against 0.160). The control reaches 0.845 TSTR-GBM where a real 300-row sample reaches 0.840: an ungrounded model exceeds real data of the same size on that student. That is the saturation this section exists to demonstrate, and it survives the prompts being matched.

A contamination control on data with no public presence. Every dataset above is a public benchmark the frozen generator has very likely seen in pretraining. So we constructed a dataset with no public presence: a renal-replacement-therapy registry of 20,000 records whose joint distribution is defined by a generative process in our own code. It has a plausible clinical schema and one deliberately counter-intuitive driver: mode of transport to dialysis outranks every clinical variable as a predictor of hospitalisation. The auto-configurator, given only the schema, independently selected `patient_age` and `transport_mode`.

We generated CoRTeC and the matched header-only control from the same release (Gemini 3.5 Flash, $n = 1,000$, two draws each). CoRTeC records 1-way TV 0.080 and TSTR-LR 0.685, against the control’s 0.541 and 0.630 and a real sample’s 0.019 and 0.693. Header-only could not reproduce the schema at all: it invented 33 levels across the four categorical columns, and 99.2% of its categorical values are absent from the real data.

The diagnostic this control specifies is whether the CoRTeC-to-header-only gap changes on data with no public presence. Against the matched control it does not: 0.057 on Adult and 0.054 here, a ratio of 0.95. CoRTeC’s margin over an ungrounded model does not shrink when the prior is removed. What the prior supplied on Adult is visible in the control’s fidelity (1-way error 0.102 on Adult against 0.541 here), not in CoRTeC’s margin.

This also answers the objection that the architecture leans on the model’s prior and would be a liability on idiosyncratic enterprise data. The prior supplies fluency in the schema, plausible values and correct types, and CoRTeC relies on that fluency; the release supplies the structure. When the prior is weak, only the release can supply it. A constructed dataset is not a real private extract, and an institutional extract remains the experiment we have not run.

6.4 Transmission: which mechanisms carry a private relationship

A generator built on a language model invites one specific doubt no aggregate metric can settle: perhaps its output is good because the model’s prior about census or clinical data happens to be right, not because it read the released statistics. We test it directly.

Definition 4 (Transmission of a private relationship). Fix a group g and a schedule of target rates $r_p, \dots, r_T$. For each r_p construct $D^{(t)}$ by resampling y within g at rate r_p , run the mechanism on $D^{(t)}$, and let $g^{(t)}$ be the realised rate of $y = 1$ within g in the synthetic output. Transmission is the pair

$$\beta = \text{slope of } g^{(t)} \text{ on } r_p, \text{ MAE} = \frac{1}{T} \sum_t |g^{(t)} - r_p|, \quad (5)$$

where r_t is the rate the mechanism was given: the released rate for a mechanism that reads a release, and the realised rate in the forced data for one fitted directly.

$\beta = 1$ with $\text{MAE} = 0$ is faithful transmission; $\beta = 0$ means the output is independent of the private data. Both must be reported, since a generator exact at the endpoints and wrong in between records $\beta \approx 1$ with a large MAE.

Table 13 shows that CoRTeC transmits, across three domains, with Stage A re-released from the modified data at every point. On NHANES the released rates were 0.008 / 0.469 / 0.970 and CoRTeC produced 0.0% / 44.6% / 100.0% over 60 to 65 rows per point. The unconditioned control is the comparison that matters in the table: it emits the same number at every target, 98.5% on hospital data against a true rate of 21.4%. CoRTeC’s output therefore tracks the released statistics rather than the model’s prior. No amount of downstream AUC reveals this, because AUC is rank-based and the direction is right, which is a concrete reason a readmission probability or a default rate must be validated on a conditional measure.

Table 13: Transmission of a forced relationship through CoRTeC. The unconditioned control emits the same number at every target.

domain	relationship forced	slope	MAE	unconditioned control	true rate
census (Adult)	education to income	1.063 [0.973, 1.168]; 3 seeds, 14 points	0.055	88.3%	61.9%
healthcare (Diabetes 130)	prior admissions to 30-day readmission	0.995	0.009	98.5%	21.4%
finance (credit)	repayment status to default	0.962 / 1.010; 2 seeds	0.019 / 0.023	100.0%	69.6%
healthcare (NHANES), auto-configured	race/ethnicity to diabetes	1.041	0.020	63.9%	15.4%

CoRTeC transmits a private relationship that contradicts the model's prior; an unconditioned LLM is flat and far from the truth

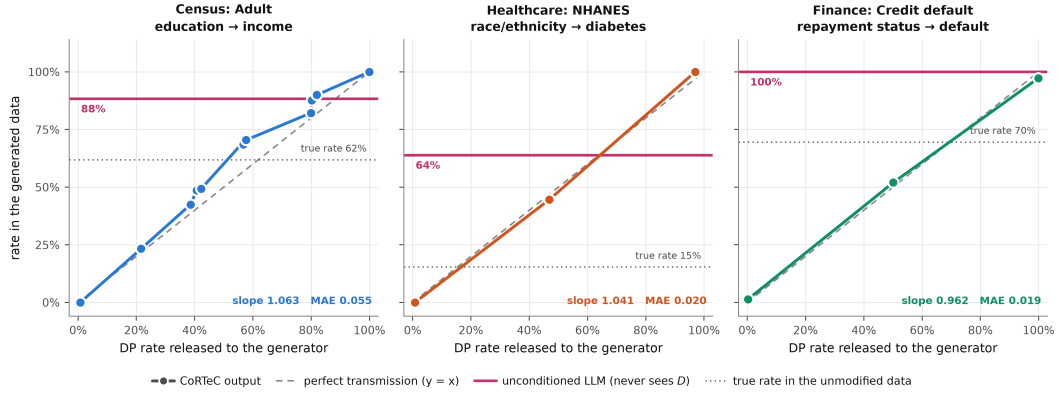


Figure 6: The transmission test. The x-axis is the released rate CoRTeC was given, the y-axis the rate it produced. The dashed diagonal is perfect transmission and the horizontal line is the unconditioned control. The healthcare panel is NHANES.

The same test on the deployed mechanisms. We ran the identical protocol on the baselines: force the rate in the private data, fit at $\epsilon = 2$ on the modified data, sample, and measure. The group is defined by public bin membership, so every mechanism can represent it. Table 14 and Figure 7 give the result, and it is not the separation we expected.

MST transmits this relationship as accurately as CoRTeC does, and the reason is structural. The manipulated quantity is a two-way marginal; MST selects and privately measures two-way marginals, and forcing the rate to an extreme makes that marginal more salient to its selection step. PATE-CTGAN does not transmit it at all. Its raw output spans only 0.079 across a target range of 1.0, the same qualitative behaviour as an unconditioned generator, reached by a different route: DP-SGD and PATE inject noise into the training signal, and at $\epsilon = 2$ on a 19-attribute schema the conditional structure does not survive it.

Table 14: Transmission by mechanism family, on identical private data at identical $\epsilon = 2$.

method	family	slope	MAE	output at 0% / 50% / 100%
MST (Diabetes 130)	marginal / graphical	0.998	0.001	0.002 / 0.506 / 1.000
CoRTeC (Diabetes 130, Claude Fable 5)	released statistics + frozen model	0.995	0.009	0.000 / 0.520 / 0.986
PATE-CTGAN (Diabetes 130)	PATE-trained generative	0.054	0.376	0.348 / 0.324 / 0.402
MST (NHANES)	marginal / graphical	0.946	0.021	0.054 / 0.500 / 1.000
CoRTeC (NHANES, Gemini 3.5 Flash)	released statistics + frozen model	1.041	0.020	0.000 / 0.446 / 1.000
PATE-CTGAN (NHANES)	PATE-trained generative	0.095	0.388	0.241 / 0.232 / 0.336

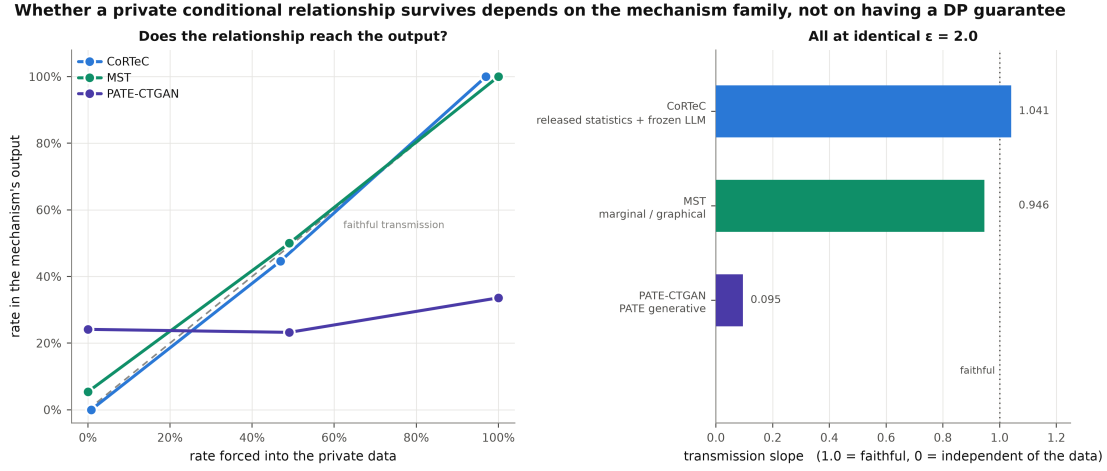


Figure 7: One forced relationship on NHANES (race/ethnicity to diabetes), three mechanisms. MST and CoRTeC track the diagonal; PATE-CTGAN’s output is very nearly flat.

Three conclusions follow.

- Whether a private conditional relationship survives depends on the mechanism family, not on whether a method carries a privacy guarantee. Two methods here are equally $\epsilon = 2$ differentially private, and one reproduces the relationship to within 0.001 while the other is effectively blind to it. A practitioner selecting on published fidelity numbers, or on the presence of a guarantee, has no way to see the distinction.
- The test establishes the narrower thing it was designed for: that CoRTeC’s output is driven by the released statistics.
- The utility gap of Section 6.1 cannot be explained by an inability to carry a conditional relationship. MST carries this one essentially perfectly and still trains models 0.15 AUC lower.

What the marginal methods lose. Ruling that out leaves the harder question of what they do lose. Table 15 decomposes each synthetic set’s departure from real data into retention of the feature-to-target association $I(X;Y)$ and of pairwise feature dependence $I(X;X)$. Each is relative to the mean over 25 real samples at the same n , so that the plug-in estimator’s small-sample bias cancels; the real-sample control scores 1.02 and 0.97, as it must.

Two distinct failure modes appear. MST and AIM inflate pairwise feature dependence by factors of 2.8 and 2.2 against a real sample of the same size, where CoRTeC inflates it by 1.24. A graphical model fitted to a set of privately measured marginals imposes exactly the dependence structure those marginals imply, and at $\epsilon = 2$ on a 15-attribute schema that structure is coarser and more strongly coupled than the data’s own. A downstream model trained on over-coupled records learns dependencies that do not hold on real test data. AIM is the instructive case: it retains the feature-to-target association almost exactly (0.98) while training models 0.17 AUC lower, so the loss cannot be attributed to the target relationship. PATE-CTGAN fails differently, retaining 37% of the association with a decision rule nearly orthogonal to the real one.

CoRTeC’s own distortion is of a different kind. It over-couples the features to the target (1.74 against 1.02), because the generator reproduces the released conditional dependence at least as sharply as the release states it. That is a calibration error a downstream user can see and correct on real held-out data, whereas over-coupled features are learned as dependencies that do not exist. It is still a real distortion, and a deployment that consumes calibrated probabilities should recalibrate before use. Across all 29 draws of seven conditions, TSTR correlates with conditional fidelity at $r = -0.64$ to -0.70 and with marginal fidelity at $|r| \leq 0.16$.

Table 15: Two failure modes (Adult, $n = 300$, earlier CoRTeC configuration). Retention is relative to a real sample of the same size.

method	$I(X;Y)$ retention	$I(X;X)$ retention	rule cosine	positive rate
real sample (control)	1.02	0.97	0.46	0.241
CoRTeC (Claude Fable 5)	1.74	1.24	0.35	0.249
AIM	0.98	2.15	0.35	0.243
MST	2.12	2.83	0.31	0.237
PATE-CTGAN	0.37	2.01	0.04	0.447

method	$I(X;Y)$ retention	$I(X;X)$ retention	rule cosine	positive rate
### 6.5 The privacy budget				

Table 16 and Figure 8 give the sweep on NHANES under the shipped configuration, one draw at each of $\epsilon = 0.3$, 1 and 8 beside the three draws of Table 5 at $\epsilon = 2$, with MST refitted at each budget. On this dataset the budget matters. NHANES has 2,999 training records, and its auto-configured release spends each cohort’s budget over ten queries and two outcome classes. At $\epsilon = 0.3$ the per-query budget is therefore 0.024, and the Laplace scale on a histogram bin is 42 records inside cohorts of 314 to 1,900. CoRTeC’s utility falls from 0.752 at $\epsilon = 2$ to 0.731 at $\epsilon = 1$ and 0.570 at $\epsilon = 0.3$, and its 1-way error rises from 0.015 to 0.063. MST degrades in the same way (0.687 to 0.587 on TSTR-LR). Between $\epsilon = 2$ and $\epsilon = 8$ nothing moves outside single-draw spread.

Table 16: The privacy-utility curve on NHANES, $n = 300$. References at this n : real-sample floor 1-way 0.034, conditional-seen 0.015, TSTR-LR 0.773; permuted-target floor 0.043 / 0.047 / 0.493.

metric	method	$\epsilon = 0.3$	$\epsilon = 1$	$\epsilon = 2$	$\epsilon = 8$
1-way TV $\downarrow$	CoRTeC	0.063	0.034	0.015	0.013
	MST	0.046	0.029	0.024	0.021
cond. TV (seen) $\downarrow$	CoRTeC	0.034	0.024	0.006	0.005
	MST	0.056	0.023	0.004	0.004
cond. TV (held-out) $\downarrow$	CoRTeC	0.045	0.027	0.036	0.035
	MST	0.116	0.143	0.128	0.137
TSTR-LR $\uparrow$	CoRTeC	0.570	0.731	0.752	0.770
	MST	0.587	0.632	0.687	0.692
TSTR-RF $\uparrow$	CoRTeC	0.606	0.729	0.732	0.699
	MST	0.575	0.628	0.646	0.668
TSTR-GBM $\uparrow$	CoRTeC	0.600	0.712	0.711	0.677
	MST	0.503	0.633	0.664	0.679

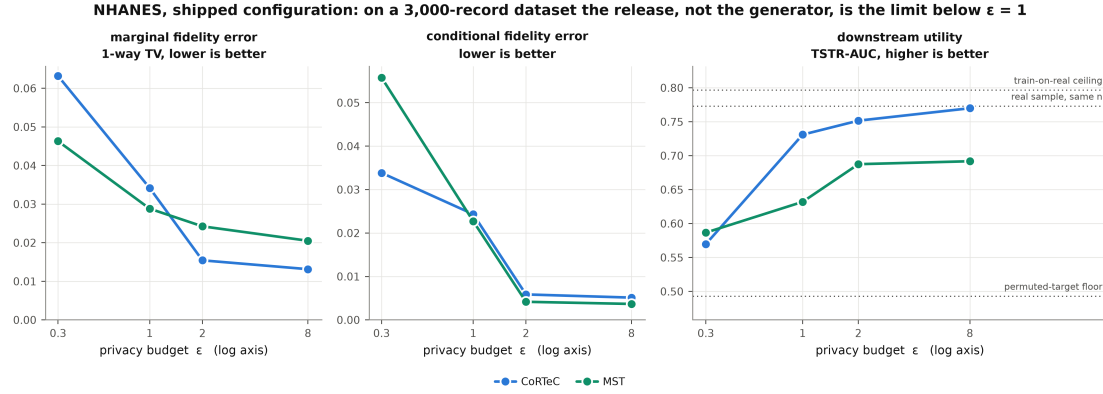


Figure 8: Privacy-utility curve on NHANES under the shipped configuration. Error panels are anchored at zero and the utility panel spans the permuted-target floor to the train-on-real ceiling, so the axes cannot exaggerate variation.

On a large dataset the curve is flat. On Diabetes 130, with 81,410 training records and the earlier configuration, CoRTeC at $\epsilon = 0.3$ is as good as at $\epsilon = 8$ on every metric except conditional error over seen groups, which degrades from 0.009 to 0.019, still better than a real 300-record sample’s 0.031. MST records lower 1-way error at every budget, by roughly 2.7 $\times$. CoRTeC’s 2-way error is lower at every budget, and its conditional error is lower by 2.2 to 7.6 $\times$. Aggregate TSTR does not separate the two at any ϵ ; both are in the range 0.54 to 0.57 against a floor of 0.557, because 30-day readmission is weakly separable.

The release itself degrades gracefully there, which is structural (Table 17). A histogram has sensitivity 1 regardless of ϵ and bin count, whereas the mean-and-standard-deviation formulation can place a released standard deviation outside its feature’s range, and there are zero such values at any budget. This can be read before any generator is run. On a small dataset the release, not the generator, is the binding constraint at tight budgets, and its noise-to-signal ratio is computable from the release alone. A regulator’s $\epsilon = 0.3$ costs almost nothing on a dataset of 80,000 records and costs a fifth of the utility on one of 3,000.

Table 17: The Diabetes 130 release across budgets (earlier configuration, 19 columns).

ϵ_{total}	ϵ per query	histogram TV vs. truth	conditional cells	max conditional error	impossible std values
0.3	0.0079	0.0492	18	0.0152	0 / 72
1.0	0.0263	0.0185	18	0.0062	0 / 72
2.0	0.0526	0.0086	18	0.0078	0 / 72

6.6 The generator

The architecture is defined by what it releases, but its output quality depends on the frozen model placed in Stage B, and that dependence is large. Table 18 measures it on Adult with full-dataset generation from one release under the earlier configuration, so the only variable is the generator.

Gemini 3.1 Pro, GPT-5, Claude Opus 5 and Claude Fable 5 fall in a band of 0.041 to 0.046 on conditional error, across three independent pretraining corpora, and every one is below what a real 300-record sample achieves (0.062). That is also the strongest available cross-corpus evidence on contamination: three separately trained models cannot have memorised Adult identically, and all track an inversion that contradicts it. Welch tests between the two Anthropic frontier models find no significant difference on any of the seven metrics (smallest $p = 0.076$). The lower block spans two vendors and three size tiers, and its common feature is that reasoning was suppressed or absent.

Table 18: Nine generator configurations on one Adult release.

generator	vendor	reasoning	draws	1-way TV ↓	2-way TV ↓	cond. seen ↓	TSTR-LR ↑
<i>train on all 26,048 real rows (ceiling)</i>			1	0.011	0.028	0.020	0.860
CoRTeC (Gemini 3.1 Pro)	Google	on	3	0.046	0.120	0.041	0.833
CoRTeC (GPT-5)	OpenAI	on	3	0.040	0.124	0.045	0.802
CoRTeC (Claude Opus 5)	Anthropic	verified off	3	0.045	0.121	0.046	0.844
CoRTeC (Claude Fable 5)	Anthropic	on	6	0.052	0.131	0.045	0.841
<i>real sample, $n = 300$</i>			3	0.043	0.107	0.062	0.830
CoRTeC (Claude Sonnet 5)	Anthropic	on	2	0.058	0.142	0.128	0.803
CoRTeC (Claude Sonnet 5)	Anthropic	suppressed	3	0.083	0.183	0.154	0.813
CoRTeC (Claude Haiku 4.5)	Anthropic	none	2	0.115	0.213	0.170	0.771
CoRTeC (GPT-5)	OpenAI	minimal	3	0.109	0.219	0.173	0.786
<i>permuted target (no-information floor)</i>			3	0.041	0.110	0.152	0.423

The controlled demonstration is GPT-5 against itself: same release, same prompts, one flag changed (Table 19). Enabling reasoning improves conditional fidelity 3.8×. The two arms are 0.128 apart on conditional error, against standard deviations of 0.032 and 0.007. Downstream utility does not resolve the difference at any feasible draw count. The direction is consistent across all three students, but resolving even the most favourable of the gaps at 80% power would take 18 draws per arm, and the least favourable 112.

The finding is that this measure cannot detect the difference. A practitioner choosing between these configurations on downstream utility would see two indistinguishable options and deploy the one that is 14.5× cheaper. They would have chosen output whose conditional structure is worse than a no-information control while it still records 95% of a real sample’s downstream AUC. Output headroom is the obvious confound, and it contributes nothing: tripling Sonnet 5’s output budget at fixed effort changes output per call not at all, and raising effort at fixed budget is the only step that moves the fidelity measures.

Table 19: One flag. Both arms use one byte-identical release and identical token limits.

GPT-5 configuration	draws	1-way TV ↓	cond. seen ↓	TSTR-LR ↑	thinking tokens	cost per draw
reasoning_effort = minimal	3	0.109 ± 0.003	0.173 ± 0.032	0.786 ± 0.050	0	\$0.19
default	3	0.040 ± 0.001	0.045 ± 0.007	0.802 ± 0.025	261k (94% of output)	\$2.81

Reasoning does not substitute for capability in either direction. Claude Sonnet 5 with reasoning verifiably firing on 34 of 34 calls reaches 0.128: a real improvement on its suppressed 0.154, but nearly three times the frontier band and only modestly clear of the no-information floor. Claude Opus 5 reaches 0.046 with reasoning verifiably off. What the band has in common is frontier-class capability. Reasoning produces a large within-model gain that a weaker model cannot convert into frontier output.

The practical guidance is:

- select a frontier-class model;
- enable reasoning, and verify that it fired from evidence that survives the transport rather than from a setting (Section 9);
- verify the output on a conditional criterion rather than on aggregate utility; and
- re-verify after any change to the generator or its configuration.

Open-weight models measured as controls transmit a forced relationship at every family we tested above 20B parameters. But on full-dataset generation a self-hosted 70B model’s output is close to indistinguishable from an unconditioned prompt on the metrics this method exists to improve. That is why the deployment proposal names enterprise platforms.

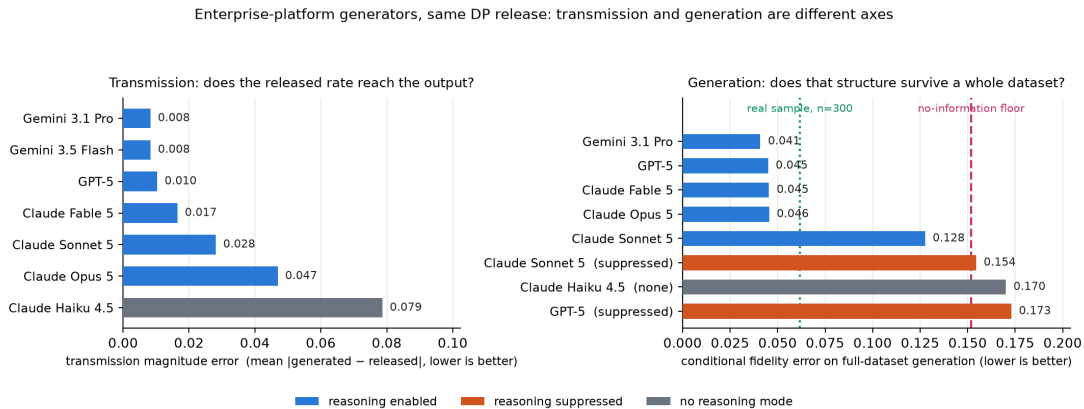


Figure 9: Enterprise-platform generators on one release. Left: transmission magnitude error on the Adult forced-relationship sweep. Every platform model is within 0.079 and the best four within 0.017, so on this axis they are nearly interchangeable. Right: conditional fidelity on full-dataset generation against the real-sample reference (0.062) and the no-information floor (0.152). The ordering changes completely, and the three suppressed or absent-reasoning arms are the ones at or past the floor.

6.7 Auto-configuration

Table 20 compares the auto-configured release against expert hand-tuning where a comparator exists, as a release-ceiling comparison at $n = 2,000$. That comparison uses real features with the target redrawn from each cell’s released rate, so it isolates the configuration and not the generator. Auto-configuration is within ± 0.013 of hand-tuning everywhere and better on one dataset. On credit it selects PAY_0 , PAY_3 and PAY_2 , the payment-history columns hand-tuning had missed and which we had independently diagnosed as a cause of that dataset’s early utility gap.

Table 20: Auto-configuration against expert hand-tuning, as a release-ceiling AUC at $n = 2,000$.

dataset	no conditioning	hand-tuned	auto-configured	real target	delta
Adult	0.655	0.825	0.812	0.858	−0.013
Diabetes 130	0.488	0.595	0.607	0.624	+0.012
credit	0.502	0.682	0.677	0.702	−0.005

It generalises to datasets it has never seen. On six additional regulated benchmarks, three health and three finance, each declaring only what a deploying institution would know without looking at its own data, the adaptive selection budget configures every one. It captures 19 to 89% of the achievable range: heart disease 68%, cervical cancer 35%, retinopathy 50%, German credit 19%, Australian credit 86% and bank marketing 89%. A fixed 5% budget declines to configure three of the six, and on one silently produces a configuration worse than not conditioning at all.

On NHANES the rule captures 87%, within two points of the best share on any dataset we tested. The columns it selected there, age, diastolic blood pressure and race/ethnicity, are genuine diabetes risk factors chosen without being told the domain, and Table 5 is generated from that release. The result holds through to generation on a second clinical dataset. On Diabetes 130 an auto-configured release generated through Gemini 3.5 Flash trains the three students at 0.609, 0.597 and 0.591; the hand-tuned release gives 0.578, 0.583 and 0.563, and a real sample of the same size 0.607, 0.587 and 0.567. The clinical equivalence there therefore holds under a configuration no human chose.

6.8 Cohort-wise or cell-wise: a representativeness failure no metric reports

Stage B can generate per cohort, as Algorithm 3 does, or per released conditional cell with an exact positive count, which expresses each cell’s rate to within one row. On a constructed registry the cell-wise path reduced conditional rate error 77×, and we shipped it as the default on that basis. On real data it failed in a way that matters more than the metric it was chosen to improve.

Table 21 compares the two paths on the same release for each dataset, byte-identical by checksum. On Adult the cell-wise path is mildly better and never worse. On NHANES it is catastrophic, and the catastrophe is invisible to every aggregate number a practitioner checks: aggregate utility reads within 0.02 of the cohort-wise path (0.752 against 0.768), while 1-way error quadruples.

Table 21: Two generation paths on one release per dataset.

dataset	metric	cohort-wise	cell-wise	Δ
Adult (2 draws)	1-way TV ↓	0.037	0.037	−0.000
Adult	cond. seen ↓	0.009	0.003	−0.006
Adult	TSTR-GBM ↑	0.847	0.877	+0.030
NHANES (1 draw)	1-way TV ↓	0.029	0.123	+0.094
NHANES	2-way TV ↓	0.131	0.260	+0.128
NHANES	cond. seen ↓	0.004	0.040	+0.036

The cause is that cell-wise generation emits rows only for released cells, so population the release does not cover is generated at rate zero. NHANES released four conditional cells, all inside a single diastolic-blood-pressure band and a single race/ethnicity group, covering 25.4% of the population; Adult’s release covered 99.9%. Table 22 shows the consequence. Four of six race/ethnicity groups, 40% of the population, are absent from the output entirely, along with every record in the 50 to 60 age band. The cohort-wise path reproduces all six groups to within 0.016 on the same release, five of the six within 0.01.

A deployment that shipped this would hold a synthetic cohort with no Asian, Mexican-American, other-Hispanic or multiracial records at all while its fidelity and utility dashboards stayed green. The conditional criterion does not catch it either, because conditional error is computed over released cells, and those are precisely the cells that are present. The model is not distorting shape within bands; it is simply never asked for the missing cells. Nothing in our own evaluation suite flagged it, and it was found by inspecting per-column output against the release.

Table 22: Race/ethnicity shares on NHANES under the two paths, from one release.

race / ethnicity	real	cohort-wise	cell-wise
White (non-Hispanic)	0.370	0.370	0.580
Black (non-Hispanic)	0.224	0.220	0.420
Asian (non-Hispanic)	0.142	0.142	0.000
Mexican-American	0.127	0.125	0.000
Other Hispanic	0.084	0.075	0.000
Other or multiracial	0.052	0.068	0.000

The fix is a per-column coverage guard, and it is free. A conditioning column is damaged precisely when the released cells fail to span it, and is otherwise untouched. Coverage is computable from marginals already in the release, so checking it is post-processing. The floor is 90% of each conditioning column’s own released mass, placed between the highest failing measurement (0.858) and the lowest passing one (0.911) across eleven per-column measurements on two datasets. An aggregate release-level rule was tried first and fails in both directions.

Below the floor the research pipeline falls back to cohort-wise generation and says so. The shipped tool refuses outright, names the safe alternative and the knobs that change coverage, and names the columns the released cells fail to span. The cohort-wise path is the default of Algorithm 3 for this reason, and it avoids the underlying trade-off. On the NHANES release above, a coarser cell-wise release with full coverage matched the cohort-wise path’s marginal fidelity, but was worse than the permuted-target floor on held-out conditional structure (0.055 against 0.053). The cohort-wise path on the richer release reached the real-sample floor on the two tree students at $n = 600$ and came within 0.026 on the linear one (0.752, 0.738, 0.735 against 0.778, 0.735, 0.725), with every group and age band present.

A second mechanism, which the guard cannot see, appeared when the rule was tested on Diabetes 130. A conditioning column the auto-configurator had coarsened into opaque groups was pinned by group membership with the within-group distribution withheld, and the model spread each group’s members uniformly. The dominant discharge code, 59.2% of real records, came out at 14.3%, one seventh of a seven-code group to three decimals, while the column’s released-band coverage scored 100% and was right to. Supplying the within-group shares, a renormalisation of proportions already in the release and therefore free, reduced the column’s error $10.7\times$ (0.641 to 0.060). The fix replicates on a second vendor’s model against the identical release, and both implementations ship it.

We present the guard as a heuristic mitigation rather than a solved component. It answers which bands of a conditioning column the released cells span, and not what the distribution is inside a band the cells did span. Establishing when a private conditional table can be conveyed to a language model without silent loss is, in our view, an open problem for this line of work. Ganev et al. [17] showed that differential privacy itself has a disparate impact on the minority subgroups of synthetic data. What we observe here is a different mechanism with the same victims: a generation path rather than the noise deletes them, and it is invisible to the same dashboards. The lesson we take is that a representativeness check over the declared schema’s categories, against the release rather than against the output alone, belongs in the standard battery.

6.9 Stage C on the primary clinical dataset

Table 23 reports Stage C on the auto-configured NHANES release at $\epsilon_{\text{cert}} = 1.0$ and $\alpha = 0.05$ over its four released cells, for the two ways the same release can be generated: per cohort, which is the shipped default, and per released cell. The ceiling clears and the floor fails in both runs, so the test discriminates and a verdict is issued each time. The two verdicts differ.

Table 23: Stage C on NHANES, one unseeded draw per condition at tolerance 0.15.

condition	worst-case simultaneous bound on the conditional gap	within tolerance 0.15?
CoRTeC synthetic, cell-wise (refused by the coverage guard)	0.0920	yes
CoRTeC synthetic, cohort-wise (the shipped default)	0.3318	no
<i>real-sample ceiling (held-out draw)</i>	<i>0.0671 (cell-wise run), 0.0745 (cohort-wise run)</i>	<i>yes</i>
<i>permuted-target floor</i>	<i>0.2076 (cell-wise run), 0.2194 (cohort-wise run)</i>	<i>no</i>

This result counts against the configuration we ship, and we state it in full. The deployable cohort-wise output, whose mean conditional error over seen groups is 0.006 in Table 5, does not clear its own transmission bound. The only output that does is the cell-wise one the coverage guard of Section 6.8 refuses, because it contains two of six race/ethnicity groups.

The two numbers are compatible. The bound is a worst-cell, simultaneous quantity, and the table’s 0.006 is a mean. In the 50 to 60-year cell, 24 of the 48 synthetic rows are positive against a private rate of 0.197, a gap of 0.30 on that one cell, and the noise term of (4) adds 0.029 at $n_{\text{min}} = 150$ and $\epsilon_{\text{cert}} = 1.0$. Cohort-wise generation pins each cohort’s marginals and shows the model the conditional table, but nothing forces the rate inside a fine cell to match it. Cell-wise generation does exactly that, which is why it clears the bound and why it is the path that drops subpopulations. A deployment that needs the bound needs a cell-wise release the guard passes, or the coverage the guard demands. A bound and a representativeness check answer different questions.

The budget must be large enough for the bound’s own control to pass, and that is measurable. At $\epsilon_{\text{cert}} = 0.5$ and tolerance 0.15 the NHANES ceiling cleared in only 21 of 40 repetitions, so a verdict issued there would have held in only about half of the runs. At $\epsilon_{\text{cert}} = 1.0$ it clears in 25 of 25 runs and the floor never does. Loosening the tolerance

instead lets the floor through too. On Diabetes 130 the procedure prints “this test did not discriminate” and issues no verdict; its worst-case bound there is driven by cells holding two synthetic rows, which are reported as thin rather than silently trusted.

The claim Stage C produces where it does clear is auditable and quantified: with probability at least 95%, simultaneously over all released cells, the private and synthetic conditional rates differ by at most 0.092. The Stage C cost is $\epsilon = 1.0$ on top of $\epsilon = 2.0$ for the release, for $\epsilon = 3.0$ per row and, on this one-row-per-person dataset, per person.

6.10 Classification profile

Membership inference is a binary classifier, and so is a model trained on synthetic data. Reporting only AUC for either hides how it behaves at an operating point. This section therefore reports both at an operating point, with the confusion matrices, on the shipped Adult draws of Table 3 (the three Gemini 3.5 Flash draws pooled, 900 records; the real-sample reference is drawn at the same size). For the attack the desirable outcome is a classifier at chance. For the downstream model the reference is not chance but a model trained on real records of the same size.

Table 24: The downstream classifier at its operating point: one tuned classifier on a classification split. The AUC column is therefore a single-model, single-split number and not a second test of Table 3’s equivalence claim.

trained on	precision	recall	F1	AUC	average precision
<i>real, full training set (26,048)</i>	0.572	0.861	0.688	0.909	0.775
<i>real sample, same n</i>	0.549	0.866	0.672	0.893	0.722
CoRTeC, shipped configuration	0.561	0.814	0.664	0.885	0.719
MST	0.514	0.720	0.600	0.784	0.512
AIM	0.392	0.612	0.478	0.710	0.399
<i>chance at base rate 0.246</i>	0.246	0.500	0.330	0.500	0.246

CoRTeC records F1 0.664 against a same-size real sample’s 0.672, within 0.008, and average precision 0.719 against 0.722. MST and AIM are 0.07 and 0.19 lower on F1. The confusion matrices show where the difference lies: a model trained on CoRTeC’s records recovers 652 of 801 positives against the real-sample model’s 694, at 511 false positives against 569.

Accuracy is excluded from the table for a reason that deserves stating, because it is the metric practitioners reach for first. Adult’s base rate is 0.246, so a classifier that predicts the majority class for every record scores 0.754, and every arm above scores between 0.67 and 0.81. In the technical report, DP-CTGAN, whose output has collapsed to a near-constant target, records an accuracy of 0.758, below the 0.759 a constant predictor gets and above AIM’s; its recall of 0.033 and F1 of 0.061 are what actually describe it. An accuracy-only comparison would rank a degenerate predictor above two genuinely useful ones. This is the same failure as in Section 6.3: a metric that cannot distinguish a degenerate predictor from a working one should not be reported alone.

The shadow-model attack, the strongest of the three cheaper attacks of Section 6.11, was run on the same 900 records at its operating point. It records accuracy 0.507, precision 0.507, recall 0.509, F1 0.508, AUC 0.511, Youden’s J 0.014 and a true-positive rate of 0.000 at a false-positive rate of 0.1%. Its confusion matrix is 505 / 495 / 491 / 509 over 1,000 members and 1,000 non-members: an even split. The same attack on the leaking control reaches AUC 0.605, accuracy 0.581, Youden’s J 0.161 and a true-positive rate of 0.037 at 0.1% false positives. Each attack metric is reported beside its chance value, because a balanced membership game has a 50% base rate, so the attack’s accuracy of 0.507 sounds meaningful and is not.

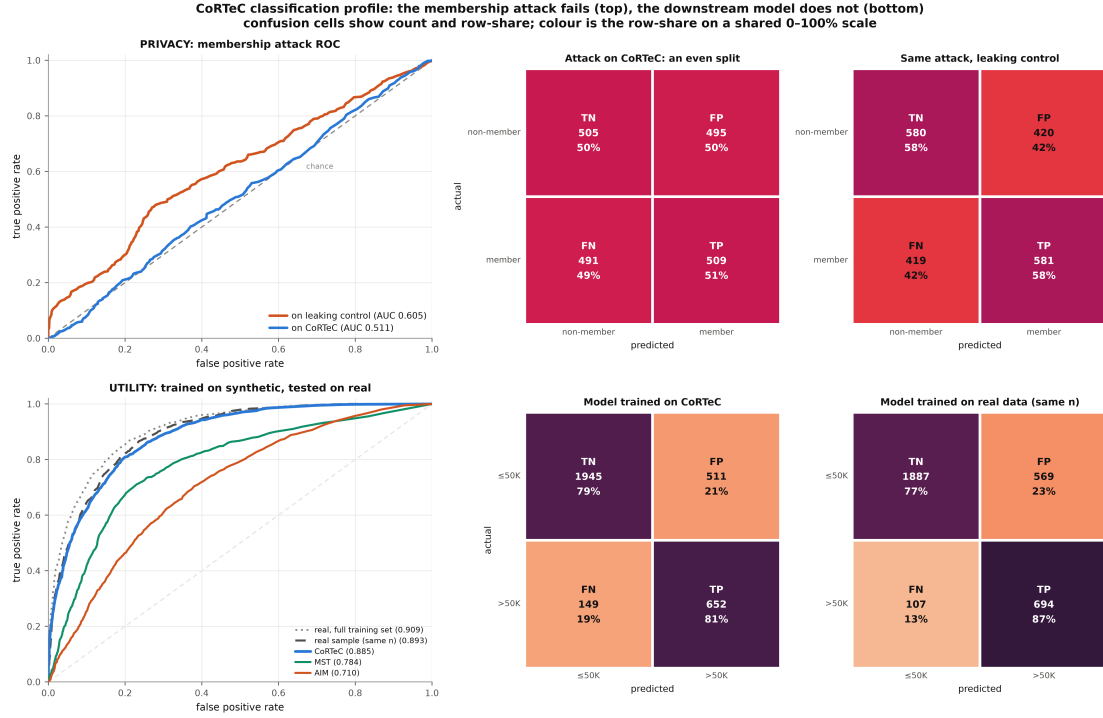


Figure 10: The classification profile on the shipped Adult draws. Top row, the privacy question: the attack’s ROC, and its confusion matrix against CoRTeC beside the same attack against a leaking control. CoRTeC’s four cells are near-identical while the control separates visibly. Bottom row, the utility question: ROC curves for models trained on each synthetic dataset, and the CoRTeC-trained model’s confusion matrix beside one trained on real data of the same size. Cells show count and row-share on a shared scale.

6.11 Membership inference

Theorem 1 already bounds any membership adversary at $\varepsilon = 2$ to an advantage of 0.762 by (2). An empirical attack cannot improve on that and cannot validate it. We run one for three reasons the proof does not cover.

- It tests the implementation rather than the mechanism. A suppression threshold not applied, a cohort of one, or a released histogram that is a delta on a single individual would each leave the theorem true and the released data leaky.
- A measured number is what security reviewers ask for.
- For a method built on a proprietary pretrained model, it is the one available answer to the objection that a private record might be memorised and reproduced. We cannot audit weights we did not train, but we can attack the artefact the method publishes, which is where such a record would have to surface to cause harm.

Setup. The standard membership game [40], in the form Stadler et al. [41] apply to synthetic data. The adversary sees the synthetic dataset and decides whether a candidate record was in the private training set. The game runs over 1,000 members from the training split and 1,000 non-members from the held-out split, disjoint and identically distributed, so the attack cannot succeed merely by recognising the data distribution. We run four attacks of increasing strength:

- nearest-neighbour distance;
- exact and near-duplicate matching;
- a discriminative shadow model trained to recognise the synthetic distribution; and
- a per-record likelihood-ratio test in the style of Carlini et al. [8]: sixteen bootstrap density models over a projection, each candidate scored by its mean log-density across shadows divided by its standard deviation, so a record the synthetic data encodes unusually consistently scores high.

Every attack is validated on a positive control, real training records passed off as synthetic, because a null from a blind attack is worthless.

Table 25: Four attacks on four datasets, with a positive control on each.

dataset	strongest advantage	exact matches	attack AUC	Youden's J	TPR at FPR 0.1%
census (Adult)	0.034	0	0.501	+0.000	0.001
healthcare (Diabetes 130)	0.017	0	0.493	−0.007	0.001
healthcare (NHANES)	0.048	0	0.524	+0.040	0.000
finance (credit)	0.046	0	0.504	+0.016	0.002
<i>chance</i>	0		0.500	0.000	0.001
<i>bound permitted at $\epsilon = 2$</i>	0.762				
<i>positive control (leaking)</i>	0.191–0.250		0.595–0.761	+0.134–0.390	0.037–0.249

On Adult the three cheaper attacks reach AUC 0.483 [0.459, 0.508], 0.498 [0.473, 0.524] and 0.500 against CoRTeC's 1,800 records, and 0.613, 0.616 and 0.625 against the leaking control, with 249 member matches and no non-member match. The per-record attack reaches AUC 0.497 against CoRTeC on the clinical data (advantage 0.006) and 0.595 against the control (advantage 0.191). The largest advantage anywhere is 0.048, against a bound of 0.762, and no synthetic record in any dataset exactly matches any private record. The positive control separates on every dataset, so the nulls are properties of the output rather than of blind attacks.

Across budgets the measured advantage stays flat where the bound grows: 0.031, 0.047, 0.028 and 0.051 at $\epsilon = 0.3, 1, 2$ and 8 against permitted advantages of 0.149, 0.462, 0.762 and 0.999. Every attack AUC's interval contains chance (Figure 11). We report the low false-positive regime as well, following Carlini et al. [8], because an average-case summary can hide an attack that is confidently right about a few individuals. The true-positive rate at 0.1% false positives is at or below 0.002 on every dataset, against 0.037 to 0.249 on the controls.

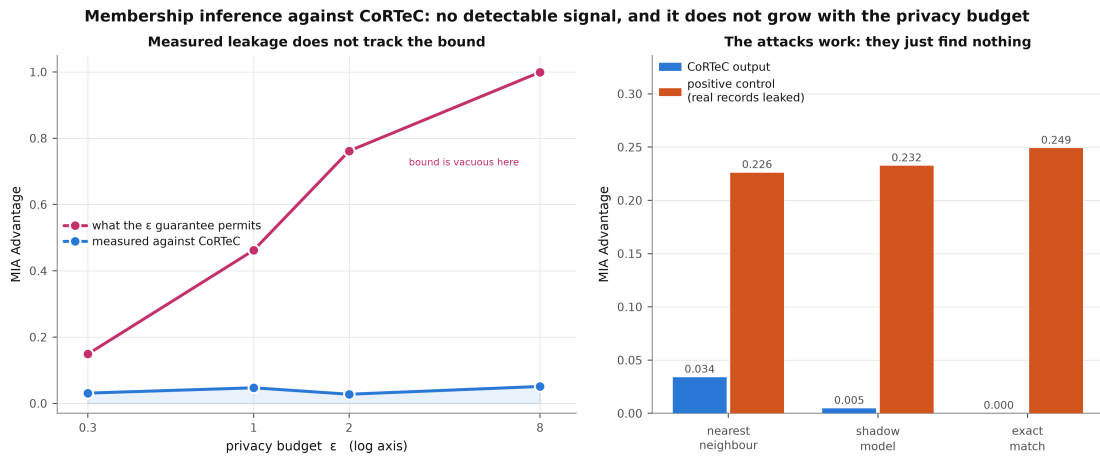


Figure 11: Left: the advantage the ϵ guarantee permits against the advantage measured, across budgets. Right: the attacks against CoRTeC beside the same attacks against a positive control.

6.12 Cost

Table 26: Computational and monetary cost. Generation costs are measured from the token counts each vendor reported over our own runs, priced at list rates. They are indicative rather than a benchmark. They exclude the reasoning-suppressed configurations of Table 18, whose apparent cheapness is not a saving.

method	fit cost	per-record generation cost	scaling
MST	23–135 s	free after fit	stable across our datasets
AIM	23–44 min (Adult); 58 s (NHANES); 16–29 min (finance, 12 columns); over 3 h without completing (Diabetes 130; finance on 15 columns)	free after fit	governed jointly by attributes and rows
PATE-CTGAN	42–65 min	free after fit	stable
DP-CTGAN	2.2 h (Adult)	free after fit	
CoRTeC, shipped configuration (Gemini 3.5 Flash)	seconds (the release)	\$1.4–2.3 per 1,000 generated rows; $\times 2$ –3 for the pool, \$4.3–7.0 per 1,000 kept records	linear in records requested
CoRTeC without the pool (Gemini 3.5 Flash / Claude Fable 5 / Gemini 3.1 Pro / GPT-5)	seconds	\$1.03 / \$4.84 / \$5.21 / \$9.36 per 1,000 records	linear

The asymmetry runs both ways. AIM’s and MST’s cost is one-off, and then sampling is free, which is decisive at the scale of millions of records. CoRTeC’s release takes seconds, its generation is trivially parallel and resumable, and it does not constrain where the compute runs. AIM’s fit did not complete within three hours on the hospital dataset or on the full finance schema. The exact-count prompt roughly doubles the reasoning tokens a call spends, and the pool multiplies the rows generated per row kept. The nine shipped-configuration draws of Section 6.1 cost \$16.6 in all.

The cheapest generator we measured is not the weakest. On the hospital data Gemini 3.5 Flash matches Claude Fable 5’s conditional error (0.010 against 0.009) with the lowest 1-way error of any CoRTeC configuration (0.035), at one fifth of the cost. It spends more reasoning per call than the Pro tier, which is consistent with Section 6.6. What the method needs is a reasoning-capable generator, not an expensive one.

7. Deployment Architecture

This section is written for the team that has to build, review and operate the system. It states where the trust boundary sits, what crosses it, and what the alternatives require. Figure 12 is the reference architecture.

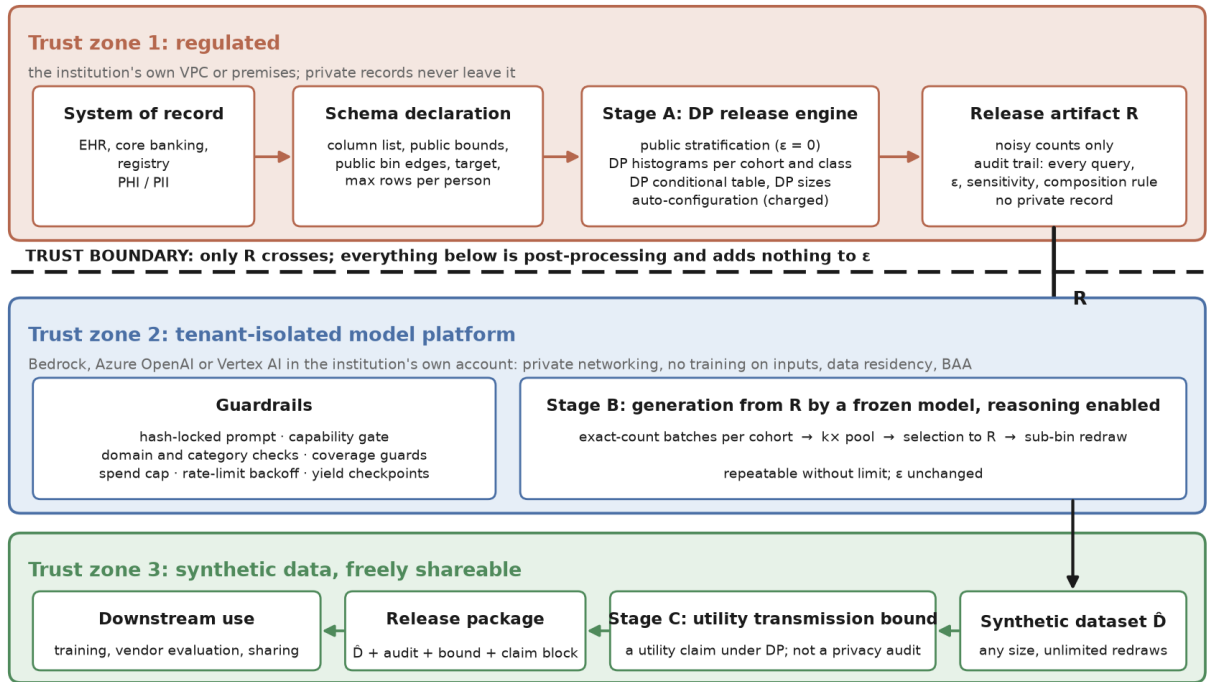


Figure 12: The reference architecture. Only one arrow crosses the trust boundary, and it carries R .

R is a differentially private release, and publishing it is what differential privacy was designed to make safe. Everything below the boundary is post-processing, so it can be repeated without limit, run by a different team, or re-run a year later, at no additional privacy cost.

Trust zone 2 is the enterprise surface, not a public API endpoint. The weights are the same either way, and CoRTeC’s privacy argument holds against a hostile endpoint by construction, since the request carries only R . What does not hold is the compliance argument, which rests on tenant isolation, private networking, data residency and a business associate agreement. Our capability results are properties of the model weights and the prompt, and the same weights are served on both surfaces. We tested that on NHANES with one release, one model and $n = 600$, three draws through the public developer API and three through Vertex AI. No metric separates the two surfaces: the largest gap is 0.006 on 2-way total variation ($p = 0.12$), and every downstream gap is at most 0.009 AUC ($p \geq 0.44$). The contractual properties were never exercised by our runs and are the deploying institution’s to verify.

Table 27: What the architecture provides, against the alternatives. The asymmetry runs both ways: the marginal methods pay once and then sample freely, which is decisive at the scale of millions of records.

	CoRTeC	DP-SGD (DP-CTGAN)	PATE (PATE-GAN, PATE-CTGAN)	MST / AIM
what crosses to the compute	private release only	raw private data	raw private data	raw private data
where compute may run	anywhere; the boundary is already crossed	training hardware inside the regulated zone	inside the regulated zone	inside the regulated zone
cost of a second dataset	zero ϵ , API cost only	zero ϵ , re-sample	zero ϵ , re-sample	zero ϵ , re-sample
cost of a changed configuration	re-release (ϵ) or reuse (free)	full refit	full refit	full refit
fit or setup time	seconds	2.2 h (Adult)	42–65 min	23–135 s (MST); 23–44 min (AIM), over 3 h without convergence on the hospital data and on the full finance schema
marginal cost per 1,000 records	\$1.03–\$9.36	free after fit	free after fit	free after fit

Controls. The release artefact carries the parameters a reviewer needs, following NIST SP 800-226 [33] and SP 800-188 [32]: the variant (pure ϵ , central, $\delta = 0$), the add/remove neighbouring relation, the privacy unit, and the composition rule and ϵ of every query. A privacy ledger is the sole source of noise scales and seals after release. A property test releases at five seeds and requires every numeric leaf of the release to move unless it is an allowlisted public quantity. That is the check that catches an undeclared query, which a ledger summing declared queries cannot (Section 9).

The measured re-identification evidence of Section 6.11 is zero exact matches on four datasets and four attacks validated on a positive control. That is the evidence that speaks to the de-identification frameworks ISO/IEC 27559 [23] and ISO/IEC 20889 [22], to Article 25 of the GDPR [16], and to the HIPAA Expert Determination route [47], which requires a statistical assessment of re-identification risk. The utility transmission bound of Stage C (Section 4.6) is deliberately excluded from that row. It bounds how much conditional structure survived generation, a utility quantity, and mapping it to a re-identification standard would be a category error with compliance consequences. The reference implementation’s bound report names the standards it does not claim, with the reason for each.

Operational failure modes. Every one of the following was observed in this project before it was guarded, and each is now enforced in code rather than documented as advice.

- A tool that re-released whenever its output directory was new quietly spent the whole budget again while producing output indistinguishable from correct. The release is now verified by hash before every generation run.
- A release that published exact cohort sizes beside correctly noised rates passed a ledger audit. The five-seed property test above catches it.
- Cell-wise generation dropped 40% of a population while every dashboard stayed green (Section 6.8). The coverage guard refuses it.
- A pool that reached its spend cap before every cohort was complete produced an unbalanced draw. The coverage guard of Algorithm 4 refuses it.
- A generator that re-typed categorical values, a transient rate limit classified as fatal, an auto-configured top band that excluded the domain maximum, and a noised cohort size clipped to zero at $\epsilon = 0.3$ were each found by a regression test that now carries the defect’s name.

Appendix C tabulates the guardrails. The tools repository carries the full operational manual.

8. Discussion

8.1 Choosing among the mechanisms

The evidence supports a selection rule stated by deliverable rather than by method.

If the deliverable is a model trained on the synthetic data, CoRTeC has the largest measured advantage on this deliverable among the mechanisms we ran. Its tree students reach the real-sample floor on every dataset; its linear student reaches it on Adult and is within 0.02 on finance and NHANES. Models trained on AIM’s and MST’s output are 0.05 to 0.18 AUC lower, and 0.14 to 0.17 lower on Adult after correction. The margin is large enough that it is unlikely to be reversed by tuning, and Section 8.2 shows that most of it is attributable to what is released.

If the deliverable is a marginal report, a contingency table or a published set of cross-tabulations, the answer depends on scale. At $n = 300$ CoRTeC’s 1-way error is within 0.005 of MST’s on Adult and finance, below it on NHANES, and below AIM’s on every dataset. At adequate n AIM records the lowest error on its own 3-way workloads, and both marginal methods sample any number of records at no cost once fitted. For a marginal report at scale, use them.

If the deliverable is a calibrated conditional rate, a readmission probability for a patient group or a default rate for a credit segment, note that it is the conditional measures rather than the marginal ones that track downstream utility ($r = -0.64$ to -0.70 against $|r| \leq 0.16$). On this axis the marginal methods’ error is furthest from the real sample’s, while their 1-way error is closest to it. A procurement decision made on 1-way total variation alone will therefore select against this deliverable.

If the constraint is scale, the marginal methods pay their cost once and then sample freely, where CoRTeC pays per record, three times over under the shipped configuration. At millions of records that favours them regardless of the utility gap.

If the constraint is that no private data may leave the institution, CoRTeC transmits a release and never a record, so the constraint is satisfied by construction even when the generator is a frontier model reached through the institution’s own tenant-isolated endpoint. DP-SGD and PATE methods place the private data on the training hardware and so constrain where they run.

If the data is small, check the release before generating. Its noise-to-signal ratio and its cell count are computable from the release at no cost, and Section 6.5 shows they say whether conditioning can contribute at all.

8.2 A hybrid for institutions that will not run a language model

Relabelling a marginal method’s output using a private conditional table is pure post-processing of two already-private artefacts, so it costs nothing beyond the table. Table 28 measures it on Adult, five base draws per method crossed with three table draws for AIM and two for MST, under the same evaluator. Relabelling moves AIM from TSTR-LR 0.675 to 0.777 and MST from 0.690 to 0.773 at no extra ϵ . At level 4 the private table matches a non-private oracle to within 0.009, so the residual is draw-to-draw noise rather than a cost paid for privacy.

Richness is necessary. A level-2 table fixes calibration and adds little utility, and the large gain arrives only with the richer table. Richer is not monotonic, since level 6 is worse than level 4 on every student for both base methods. This is also the strongest available answer to the objection that the marginal methods are merely under-tuned: the gap is largely attributable to what each mechanism releases, and a practitioner who wants the conditional structure can obtain much of it by augmenting the marginal method.

Assignment is by an exact per-cell count with stochastic rounding over a random permutation of the cell. A rank-preserving assignment, our original default, was measured to be worse on three datasets at three seeds each (independent assignment within a cell scored higher in 9 of 9 runs). The guard meant to enable ranking scored the synthesiser’s self-consistency rather than its validity, and so selected for the failure it was built to prevent.

The hybrid ships as the second reference implementation (`cortec-hybrid`). It reports whether the correction helped and tells the user to keep their budget when it did not. Across eleven datasets the correction improves conditional calibration almost everywhere while moving downstream AUC in both directions.

Table 28: The hybrid on Adult. Giving AIM and MST the quantity neither releases recovers most of the distance to CoRTeC without altering either synthesiser’s configuration.

condition	1-way TV ↓	cond. seen ↓	TSTR-LR ↑	TSTR-RF ↑	TSTR-GBM ↑
AIM alone	0.037	0.054	0.675	0.700	0.691
AIM + private conditional table, level 2	0.037	0.014	0.700	0.726	0.680
AIM + private conditional table, level 4	0.038	0.054	0.777	0.840	0.810
AIM + level 4, non-private oracle table	0.039	0.053	0.786	0.841	0.810

condition	1-way TV ↓	cond. seen ↓	TSTR-LR ↑	TSTR-RF ↑	TSTR-GBM ↑
AIM + level 6	0.038	0.071	0.742	0.778	0.735
MST alone	0.024	0.169	0.690	0.728	0.681
MST + level 2	0.025	0.053	0.727	0.737	0.695
MST + level 4	0.025	0.094	0.773	0.838	0.807
MST + level 4, non-private oracle table	0.025	0.096	0.780	0.843	0.815
MST + level 6	0.025	0.118	0.717	0.757	0.716

9. Threats to Validity: Measurement Artefacts

Defects arose throughout this work, and every one of them produced a plausible but wrong scientific conclusion before it was caught. We report them because we believe the field under-reports them, and because each is a trap for any group building a comparable pipeline. The technical report tabulates twenty-seven that bear on measurement and privacy accounting. The ones a reader should weigh are these.

The instruments, not the results. A released quantity that carries no noise is invisible to a budget audit built by instrumenting the noise mechanism. Our research pipeline published per-cohort record counts exact for most of the project while the audit reported a clean ϵ throughout, because the audit summed the queries it could see. The shipped tool had earlier been caught publishing exact cohort sizes and per-cell supports beside correctly noised rates, a real violation, again with a clean ledger.

The corrected check is behavioural rather than structural: re-release the same data at several seeds and require that every published quantity moves unless it is explicitly public. It found the exact counts immediately, together with a rule that had been deleting rare categories from releases and, with them, real subpopulations (`race = Amer-Indian-Eskimo` from a census cohort, `race_ethnicity = other_multi` from a health-survey cohort). Neither changed a reported fidelity or utility number, because the counts were accurate and the categories were rare. Both changed what the release was entitled to claim about itself, which is the more serious kind of error. A verifier that sums declared queries cannot detect an undeclared one.

A saturated metric. The original primary metric of this project scored the real-data ceiling and the unconditioned baseline identically, and a verdict had been drawn from it before anyone compared the two. Checking every metric’s floor against its ceiling before trusting it caught that defect, and later a released standard deviation of 693.80 on an attribute whose range is a few decades. It produced Section 6.3.

The defect with a named victim. Every other entry corrupts a number. Cell-wise generation removed people (Section 6.8). The synthetic dataset was internally consistent, passed its fidelity checks and trained models to within one point of the alternative, and it contained no Asian, Mexican-American, other-Hispanic or multiracial records at all. Nothing in our own evaluation suite flagged it, including the conditional criterion this paper advances, because conditional error is computed over released cells, and those were the cells that were present.

The privacy claim, not a measurement. A tool that re-released whenever its output directory was new quietly performed a second release, spending a full ϵ_{total} again on the same data, while producing output indistinguishable from correct. The mechanism’s headline property, that any number of datasets may be drawn from one release at no further cost, was violated by an implementation detail. We now verify the release artefact by hash before every generation run.

Telemetry needs a floor-and-ceiling check too. A count that was never reported is not a count of zero, and an instrument that conflates the two converts missing data into evidence for whichever conclusion that zero supports. Two published claims rested on such zeros. One was that a model’s reasoning was “off” in an arm that was in fact running at high effort. The other was that another model’s reasoning was intermittent, when the two batches compared differed in transport and only one of them was measured. Both were withdrawn once the presence of a reasoning block, which is visible on every transport, replaced the token count as the primary evidence. Before trusting a measurement, establish that it can distinguish the two states you are asking it about, and apply that to the telemetry as well as to the results.

A repaired defect can return through a dependency upgrade. A library changed how a missing value renders, months later, in code we do not control. A repair that worked and was tested silently stopped firing. No test failed, and the corruption was invisible at the call site. It surfaced only because a published table was re-run in a second

environment and one baseline moved from 0.019 to 0.110. A separate loader defect had labelled a missing HbA1c measurement as a negative case, because `NaN ≥ 6.5` evaluates to false and the resulting string is not null; 135 of the NHANES respondents were mislabelled until an assertion on the target’s provenance caught it. We pin the analysis environment, state the pin, and treat a missing-value convention as an assumption that deserves an assertion.

A binning convention. The evaluator bins with right-closed intervals while the release, the selection and the baselines’ discretisation are left-closed, so an integer on a bin edge counts differently under the two (Section 5.3). It was found while adopting the sub-bin rule, whose bin invariance holds under the release’s convention and not the evaluator’s. Every published number is the evaluator’s, and the ordering of methods is the same under either.

Four practices caught these, and we recommend all four.

- Check every metric’s floor against its ceiling before trusting it.
- Report several student models. This caught a hybrid result that reversed sign between linear and tree students.
- Inspect raw responses before concluding a backend is incapable. A truncated prompt, an output budget consumed by hidden reasoning and a mis-set effort parameter are indistinguishable from a negative result at the level of summary statistics.
- Check that a check can fail. Three separate verifications in this project gave false assurance: a ledger that summed only declared queries, an acceptance check whose name promised more than its assertion delivered, and a test module that had never run because of an import error.

10. Limitations

1. **The head-to-head is at $n = 300$ under one generator family.** The shipped configuration is measured with Gemini 3.5 Flash at three draws per dataset. The cross-vendor check through Claude Opus 5 is one complete pool on Adult, and the powered six-draw equivalence test was run with Claude Fable 5 on the earlier release design. Three draws under a second vendor, and a second vendor on finance and NHANES, are the replications the result still needs. At $n = 1,000$ on Adult the point estimates favour the real sample, and we state the equivalence claim at the size where it was demonstrated with adequate power.
2. **AIM enters the finance comparison only on a reduced schema and is absent from the hospital data.** Its fit exceeded three hours on the full 15-column finance schema and on Diabetes 130, against its own paper’s 24-hour allowance. The cause is localised to the six high-cardinality amount columns. On NHANES it completes in under a minute, and Table 5 reports it there.
3. **On NHANES the linear student and 2-way error do not reach the floor.** Logistic regression trains 0.021 short ($p = 0.047$), and 2-way error is 0.119 against the sample’s 0.081. The remaining linear gap lies in the one cohort whose diabetic class does not clear $n_{\min}$, where the output follows the generator’s placement below bin resolution rather than the release’s. Finer public bins where cohort sizes support them would leave the generator less room; sizing them is an auto-configuration rule we have not built.
4. **The suppression decisions are uncharged.** Which cohorts and cells appear, and hence how many levels share ϵ_{cond} , depend on the data. We follow the marginal-synthesis literature in treating them as released, so that our $\epsilon = 2$ rows are comparable to MST’s and AIM’s. The reference implementation is more conservative than this paper’s tables in that respect. The earlier-configuration arms additionally carry the (ϵ_L, δ) guarantee of Section 4.7 rather than pure ϵ .
5. **ϵ protects a row.** On encounter-level data with repeated individuals the per-person guarantee is weaker by the maximum contribution count, 40 on Diabetes 130. The auto-configurator has no concept of a person; on that dataset it selected `number_inpatient`, a proxy for how many rows a patient contributes. MST, compared against us at row-level ϵ on the same data, inherits the same per-person figure. The mitigation is the contribution cap of Section 3, taken before the release.
6. **Every real dataset is a public benchmark the generator has very likely seen.** The constructed registry of Section 6.3 rules out gross recitation and shows CoRTeC’s margin over an ungrounded model unchanged when the prior is removed. It does not rule out recall of the marginal distributions. An institutional extract, or a model of known training provenance, remains the experiment we have not run.

7. **The membership-inference result is bounded by the attacks we ran.** An absence of detectable leakage under four attacks validated on a positive control is not an absence of leakage. Adversaries stronger than these are excluded by the ϵ guarantee, not by our measurement.
8. **The transmission sweeps are single-seed on the clinical datasets.** Each perturbs one relationship at three interior points, so the curve shape between endpoints rests on one measurement per model.
9. **Quality is bounded by the generator and by its configuration.** A single flag moved conditional error 3.8 $\times$, and the flag must be verified from the vendor's reported reasoning evidence rather than from a setting. Open-weight models we measured as controls did not reach the frontier band, which is why the deployment proposal names enterprise platforms.
10. **The coverage guard is a heuristic.** Its 90% floor lies in a five-point gap between the highest failing and lowest passing measurements. It answers which bands were released and not what is inside one. When a private conditional table can be conveyed to a language model without silent loss is open.
11. **Per-record cost** is decisive against CoRTeC at the scale of millions of records, and the shipped configuration generates three rows for every one it keeps.

11. Conclusion

The question a practitioner asks of synthetic data is simple: if I train on this instead of on the real thing, how much accuracy do I lose? For CoRTeC at $\epsilon = 2$ and $n = 300$ on three datasets, the answer is nothing that two tree students can detect on any dataset, and at most 0.02 AUC for a linear student on two of the three. The two marginal mechanisms in deployment today lose 0.05 to 0.18 AUC over the same data. CoRTeC's own marginal error is within 0.005 of the most accurate of them on Adult and finance and below it on NHANES, below a real sample of the same size on all three, at the release's own distance from the private data.

The mechanism trains nothing. It spends the budget once, on cohort-conditioned histograms per outcome class and a conditional table over a disjoint partition, where parallel composition makes conditional structure cheap. It lets a frozen language model decode the release under exact counts, selection and a sub-bin rule, which together reach the release's own fidelity at no privacy cost. Because only the release crosses the trust boundary, the generator can be a tenant-isolated enterprise endpoint, and the choice of model becomes a quality decision rather than a compliance one.

Two of the findings reach beyond the mechanism. Both criteria the field selects synthesisers on are saturated. Real data with its target permuted passes a 90% marginal-similarity bar, and on a public benchmark an ungrounded language model with no access to the private data comes within 0.010 of a real sample and exceeds the deployed mechanisms on aggregate downstream utility. Yet it is wrong about 72% of the subgroup where the institution's data departs from public knowledge, and it reports a 98.5% readmission rate for a group whose true rate is 21.4%. Only conditional and subgroup measures, reported against a real-sample floor and a no-information floor, separate a mechanism that transmits an institution's conditional structure from one that reproduces what was already public.

And whether a private relationship survives synthesis is a property of the mechanism family, not of the guarantee. MST carries a forced two-way relationship as faithfully as CoRTeC does and still trains models 0.15 AUC lower, because what it loses is measurable as a 2.8 $\times$ inflation of pairwise feature dependence.

Where CoRTeC belongs is narrow and, we think, useful. It is the mechanism to reach for when a downstream model or a calibrated conditional rate is the product, and not the mechanism to reach for when a marginal report at scale is.

Ethics Statement

This work uses public benchmark datasets and a national health survey that is released for research. No private institutional data was used. The membership-inference attacks were run against synthetic data we generated from those benchmarks, with the positive controls constructed from the same public records. The mechanism is intended to reduce the privacy risk of sharing tabular data, and we have tried to state its guarantee and its limits precisely. The row-level privacy unit, the uncharged suppression decisions, the pretraining corpus outside the guarantee, and the difference between a utility bound and a privacy audit are each stated where they arise. A reader who takes any of them for more than it is would be worse protected than they believe.

Reproducibility Statement

Two repositories accompany this paper. The research repository ([Calyie/cortec](#)) holds the pipeline that produced every table, the figure and build scripts, and the companion [technical report](#). The per-draw result records every published number is re-derived from, and the audit suite that re-derives them, are retained in the project’s internal tree and are available from the authors on request. The audit suite has four scripts. One recomputes every headline number from the stored records. One asserts that the paper and the shipped tools agree on every parameter and rule. One scans the code for the defect patterns of Section 9. One checks that every number in this paper appears in the technical report.

The tools repository ([Calyie/cortec-framework](#)) holds the two Apache-2.0 reference implementations with a test suite in which every regression test is named after the defect it prevents. Appendix D gives the environment, the commands and the spend. Released statistics cannot be regenerated bit for bit, because the noise source is cryptographically secure and ignores any seed; every stored release is therefore kept, and every draw reuses it by file.

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Appendix A. The Prompt

The reference implementation freezes and fingerprints its templates, verifies the fingerprint before every render, and requires a user who needs a different template to register it under a new version. It does so because two elements are essential and their removal is not visible in the output: the pipeline still runs and still produces plausible rows, and is simply no longer conditioned on the data. The system message and the cohort template follow, with the exact-count block that Algorithm 3 fills.

```

1  SYSTEM
2  You generate synthetic tabular records for privacy research. You will be
3  given summary statistics that were computed under differential privacy.
4  Reproduce the distributions and relationships they describe as closely as
5  you can, EVEN IF THIS CONTRADICTS YOUR EXPECTATIONS about how these
6  variables usually relate. The statistics describe this specific
7  population; your general knowledge does not. Output only CSV rows. No
8  commentary, no explanation, no markdown fences.
9
10 USER
11 Generate exactly {n_rows} synthetic records for the following subgroup.
12
13 SUBGROUP: {cohort_name}
14
15 These statistics were measured on the real data for THIS subgroup, under
16 differential privacy. Match them. Where a statistic below conflicts with
17 what you would otherwise expect, follow the statistic: it is a
18 measurement of the actual population and your expectation is not.
19
20 NUMERIC DISTRIBUTIONS (released histograms; reproduce the shape, not just
21 the mean)
22 {numeric_block}
23
24 CATEGORICAL DISTRIBUTIONS (all released categories and their proportions)
25 {categorical_block}
26
27 OUTCOME RATE IN THIS SUBGROUP
28 {class_balance_block}
29
30 CONDITIONAL OUTCOME TABLE (P({target} = {positive} | cell), measured under
31 differential privacy)
32 This table is the most important part of this prompt. Each row states the
33 outcome rate for a specific cell of the population. Your generated records
34 must reproduce these rates, even where a rate is the opposite of what you
35 would expect for that group.
36 {conditional_block}
37
38 EXACT COUNTS FOR THIS BATCH (the number of rows, out of {n_rows}, that
39 must fall in each bin, each category and each outcome; these are the
40 counts this subgroup still owes)
41 {counts_block}
42
43 OUTPUT FORMAT
44 Exactly {n_rows} CSV data rows, plus one header row, with these columns in
45 this order:
46 {columns}
47
48 Rules:
49 - every numeric value must lie within its stated range
50 - {target} must be exactly "{positive}" or "{negative}"
51 - do not repeat identical rows; vary records the way real data varies
52 - output nothing except the header and the {n_rows} data rows

```

In a cohort with class-conditional blocks, the numeric and categorical blocks are given once per outcome class. The header-only control of Section 6.3 is this prompt with the five released blocks removed and nothing else changed. Every value in every block is a released quantity; the prompt never carries a private record.

Appendix B. The Stage C Report

Stage C (Section 4.6), shipped as `cortec.bound` in the framework and as `certify.py` in the research repository, writes its result as a report a third party can read without the private data. The report is headed `UTILITY TRANSMISSION BOUND`. Its JSON:

- opens with a `_what_this_is` block stating that it is not a privacy audit;
- records a clearing result in a field named `within_bound` and never `certified`;
- lists every released cell with its bound and its synthetic support, so that thin cells are visible;
- states ϵ_{cert} , α , the tolerance, the privacy unit and the total ϵ spent by release and bound together;
- claims alignment with NIST SP 800-226 and SP 800-188 only; and
- carries a `_standards_not_claimed` block naming HIPAA Expert Determination, ISO/IEC 27559, ISO/IEC 20889 and GDPR Article 25 with the reason for each.

The bound is unseeded, because it spends budget on a query over the private data and a seeded mechanism is deterministic. A published bound is therefore one draw. The operating point ($\epsilon_{\text{cert}} = 1.0$, tolerance 0.15) was chosen by repetition rather than from a single run (Section 6.9). A Monte Carlo over 200 independent noise draws checks that the empirical violation rate of (4) stays below α .

Appendix C. Implementation Notes for Deployment

This appendix carries what an implementer needs from the paper. The operational manual, with the full failure-mode catalogue, the deployment checklist with its rationale, the two deployment patterns, the surface adapters and the cost model, ships as [documentation in the tools repository](#).

C.1 Parameters. Table 29 lists the shipped defaults and what each was set from.

Table 29: Shipped parameters.

parameter	default	set from
ϵ_{total}	2.0	the budget every head-to-head in this paper runs at; on a large dataset the output is flat over [0.3, 8], on a small one it is not (Section 6.5)
conditional share α	0.2	a sweep of the release decoded with no model: a fifth costs no measurable conditional accuracy once the release carries class blocks, and the histograms' error falls by a third
count share γ	0.02 (research pipeline), 0.05 (tool)	the published cohort size is a private count and is paid for out of the marginal share
suppression floor n_{min}	150	a conditional rate over fewer than about 50 records is dominated by its own Laplace noise; 150 leaves the class blocks estimable on cohorts of a few hundred; the tool refuses $n_{\text{min}} < 50$
batch size B	25 rows	the largest batch on which the frontier models reproduce every count line exactly
pool factor k	3 (2 on NHANES)	the selection's gain saturates between 2 $\times$ and 3 $\times$; each further unit costs one generation spend
pool coverage floor	1.1 $\times$ the rows each cohort owes	the failed pool of Section 4.5
cell-wise coverage floor	90% of each conditioning column's released mass	placed between the highest failing measurement (0.858) and the lowest passing one (0.911) over eleven per-column measurements on two datasets
sub-bin rule	<code>release</code> (opt-out generator)	Table 10
reasoning	<code>on</code>	Table 19; suppressing it warns with the measured 3.8 $\times$ cost

C.2 The release artefact. The release is a JSON document containing no private record. Per cohort it carries the published size, the class balance, and one histogram per attribute and per outcome class, or one pooled histogram where a class fell below n_{min} . Per conditional level it carries each surviving cell's published positive count, size and rate. Its audit block records every query with its ϵ , sensitivity, composition rule and partition key, and the totals: `epsilon_accounted` (per row), the privacy unit with `max_rows_per_person` and `epsilon_per_person`, the variant (pure ϵ , central, $\delta = 0$), and the neighbouring relation (add/remove). A documented-gaps block names the floating-point Laplace mechanism, the pretraining provenance of the generator and the uncharged suppression decisions.

The ledger seals after release, and any later attempt to spend raises. The artefact cannot be regenerated, because the noise source is cryptographically secure and ignores any seed. The stored artefact is therefore the only copy of that release, every draw must reuse it by file, and the generator verifies it by hash before every run.

C.3 Guardrails enforced in code. Each row of Table 30 corresponds to a failure observed in this project, and each is a regression test named after it.

Table 30: Guardrails.

failure	guardrail
a second release fired silently, re-spending the budget	release verified by hash before generation; a fresh release prints a banner and requires confirmation
a released quantity published exact beside noised ones	property test: release at five seeds, every numeric leaf must move unless allowlisted as public
generator below the capability floor	capability gate: measured-insufficient models refused, unmeasured ones require an explicit override
reasoning suppressed or not verifiably fired	reasoning="on" default; the presence of a reasoning block is recorded per call and an unreported count is surfaced as unverified, not as zero
context truncation cutting off the conditional table	context-fit check before the call
out-of-domain values, undeclared categories, re-typed categoricals	domain-bounds rejection; the parser normalises to the declared strings and drops undeclared categories, counting them
whole subpopulations absent under cell-wise generation	per-column coverage guard at 90%; the tool refuses and names the column, the safe alternative and the knob
a pool that stopped early	pool-coverage guard at 1.1x; selection refuses
a noised cohort size clipped to zero	published size clamped at $n_{\min}$
stratification bands with a gap at the domain maximum	bands must tile the domain; the top band is closed
a transient rate limit counted as a failure	exponential backoff (15 s, doubling, six waits) before a rate-limited call counts against the yield rule; billing and credential faults stay fatal
spend runaway	per-run cap, parse and yield checkpoints every two calls, partial output preserved
a vacuous privacy unit on encounter-level data	the schema must declare <code>max_rows_per_person</code> ; a vacuous ϵ_{person} is refused unless acknowledged in the audit trail

C.4 Deployment checklist, condensed.

1. Declare the schema from public knowledge only: column list, domain bounds, bin edges from clinical or regulatory convention, the target.
2. Establish the privacy unit. On encounter-level data, cap each person’s contribution to their first C rows before Stage A, choosing the smallest C whose $\epsilon_{\text{person}} = C \cdot \epsilon_{\text{row}}$ the regulator will accept, and re-check cell coverage afterwards.
3. Choose ϵ and n . On a small dataset, inspect the release’s noise-to-signal ratio before generating.
4. Run Stage A once; store and hash the artefact.
5. Check the release: whether the finest conditional table is non-empty, how many cells cleared $n_{\min}$, and what share of each conditioning column’s mass the surviving cells span.
6. Select a reasoning-capable frontier model on a tenant-isolated enterprise platform in the institution’s own account, enable reasoning, and verify that it fired.
7. Generate, reusing the artefact. Draws are free in ϵ .
8. Evaluate against a real sample at matched n and the same data with its target permuted, fidelity and utility in one table, and compare each declared categorical’s full support against the release.
9. Compute the transmission bound and attach it, the audit trail and the claim block to the release package.
10. Re-verify after any change to the generator or its configuration.

Appendix D. Reproducibility Details

Artefacts. The research repository holds the pipeline that produced every table: Stage A and the prompts in `src/generic_pipeline.py`, the backends in `src/llm_generator.py`, the auto-configurator in `src/autoconfig.py`, the driver `run_dataset.py`, the baselines runner, the evaluator, the transmission and

membership-inference scripts and the statistics; and the figure and build scripts. The per-draw result records and the audit suite named in the reproducibility statement are internal and available on request. The tools repository holds the two Apache-2.0 reference implementations, `cortec` (Figure 2) and `cortec-hybrid` (Section 8.2). Test counts at the time of writing: 136 in `cortec` and 26 in `cortec-hybrid`. Every regression test was verified to fail against the pre-fix code.

Regenerating an arm. Every CoRTeC arm is one invocation of the driver. The shipped Adult arm of Table 3 is

```
1 python3 run_dataset.py --dataset adult --stage generate --backend gemini --surface vertex \  
2   --model gemini-3.5-flash --n-synthetic 300 --n-out 300 --draws 3 --rows-per-call 25 \  
3   --epsilon-total 2.0 --cond-frac 0.2 --n-min 150 --pool-factor 3 --seed 42 \  
4   --skip-header-only --no-by-cell --outdir results/adult_v2_gemini
```

A release already present in the output directory is reused rather than redrawn. Every table is scored by `evaluate_generic.py` at seed 42, and the technical report records the exact arm specification behind each table.

Environment. Every table is produced under pandas 2.3.3 with scikit-learn 1.6.1. pandas 3.0.3 with scikit-learn 1.9.0 agrees to within 0.002 on every cell; the residual is a tree-seeding difference. scikit-learn is pinned below 1.9 for compatibility with diffprivlib [21], whose Laplace mechanism we use with a verified from-scratch reimplementaion as a fallback. Baselines use smartnoise-synth 1.0.8 [34].

Privacy accounting is verified by instrumentation: a test patches the Laplace mechanism inside the release path and tallies what is spent, asserting exactly ϵ_{total} on three auto-configured schemas and at three budgets, with per-cohort query counts adapting to 15-, 10- and 19-column schemas.

Spend. Total commercial spend for every experiment in this paper and the technical report was approximately \$220 as metered. We report it as an estimate: one vendor bills reasoning tokens separately from output tokens, and a price-table prefix match once charged a cheap model at a frontier model’s rate. Report tokens, and treat any dollar figure as an estimate.